

System Programming

Linux / UNIX

Basic assignment

학 과: 컴퓨터정보공학부

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학 번: 2020202090

성 명: 최민석

1. Basic-1 Installation

- VMware installation

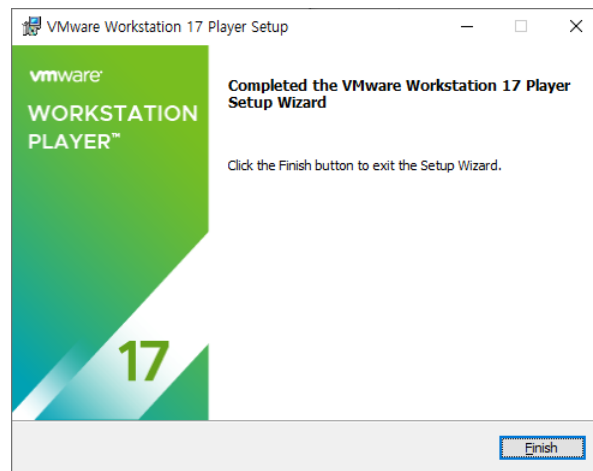
1. VMware 17.5 설치 파일을 다운로드

▼ 오늘 (2)			
 미확인 92550.crdownload	2024-03-21 오후 8:12	CRDOWNLOAD ...	16KB
 VMware-player-full-17.5.0-22583795.exe	2024-03-21 오후 8:13	응용 프로그램	553,153KB

2. 설치 파일 실행



3. 설치 완료



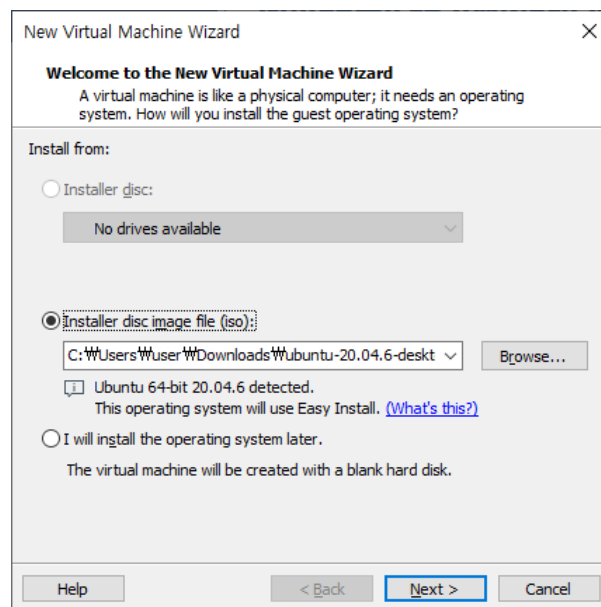
- Ubuntu installation

1. Ubuntu 20.04.6 설치용 ISO 파일 다운로드

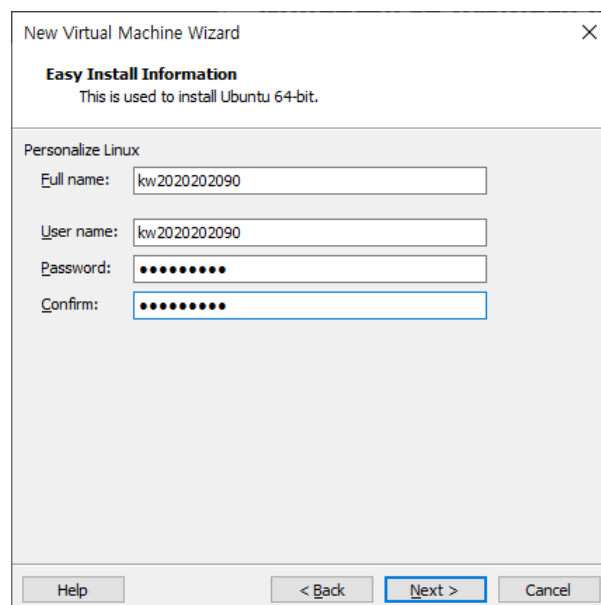
▼ 오늘 (2)

VMware-player-full-17.5.0-22583795.exe	2024-03-21 오후 8:13	응용 프로그램	553,153KB
ubuntu-20.04.6-desktop-amd64.iso	2024-03-21 오후 8:21	디스크 이미지 파일	4,249,476...

2. VMware에서 새 가상머신 생성



여기서 installer disc image file에 아까 받은 ubuntu iso 파일을 등록해준다.



사용자 이름과 비밀번호를 설정한다.

New Virtual Machine Wizard

Name the Virtual Machine
What name would you like to use for this virtual machine?

Virtual machine name:

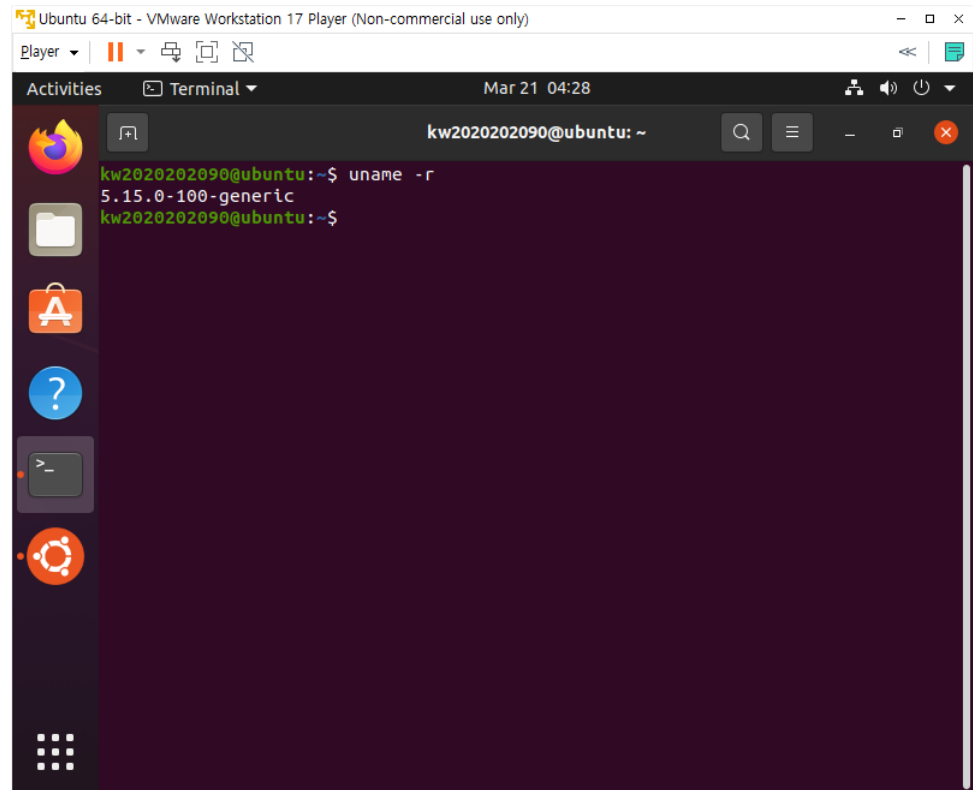
Location:

< Back **Next >** Cancel

가상머신의 이름을 설정하면 생성이 완료된다.

3. Ubuntu 설치 (\$uname -r)

이후 생성된 가상머신을 실행하면 ubuntu가 자동으로 설치된다.

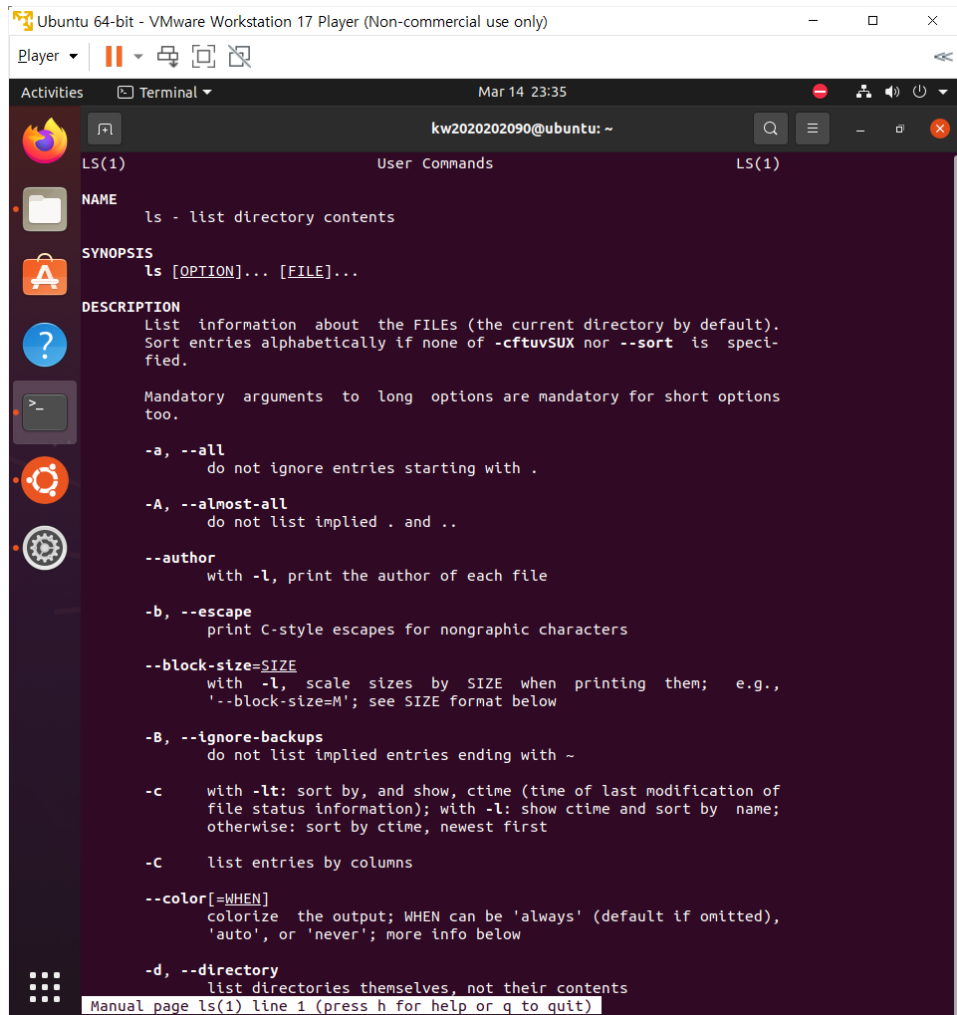


설치 완료

추가적으로 VMware에서 가상머신 생성시 AMD Ryzen CPU는 BIOS 환경에서 SVM Mode를 활성화시켜줘야 한다.

2. Basic-2 Linux Commands

- man
- man ls



The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal displays the manual page for the 'ls' command. The window has a dark theme with a purple background. The terminal output is as follows:

```
LS(1)                                User Commands                                LS(1)

NAME
  ls - list directory contents

SYNOPSIS
  ls [OPTION]... [FILE]...

DESCRIPTION
  List information about the FILES (the current directory by default).
  Sort entries alphabetically if none of -cftuvSUX nor --sort is speci-
  fied.

  Mandatory arguments to long options are mandatory for short options
  too.

  -a, --all
      do not ignore entries starting with .

  -A, --almost-all
      do not list implied . and ..

  --author
      with -l, print the author of each file

  -b, --escape
      print C-style escapes for nongraphic characters

  --block-size=SIZE
      with -l, scale sizes by SIZE when printing them; e.g.,
      '--block-size=M'; see SIZE format below

  -B, --ignore-backups
      do not list implied entries ending with ~

  -c
      with -lt: sort by, and show, ctime (time of last modification of
      file status information); with -l: show ctime and sort by name;
      otherwise: sort by ctime, newest first

  -C
      list entries by columns

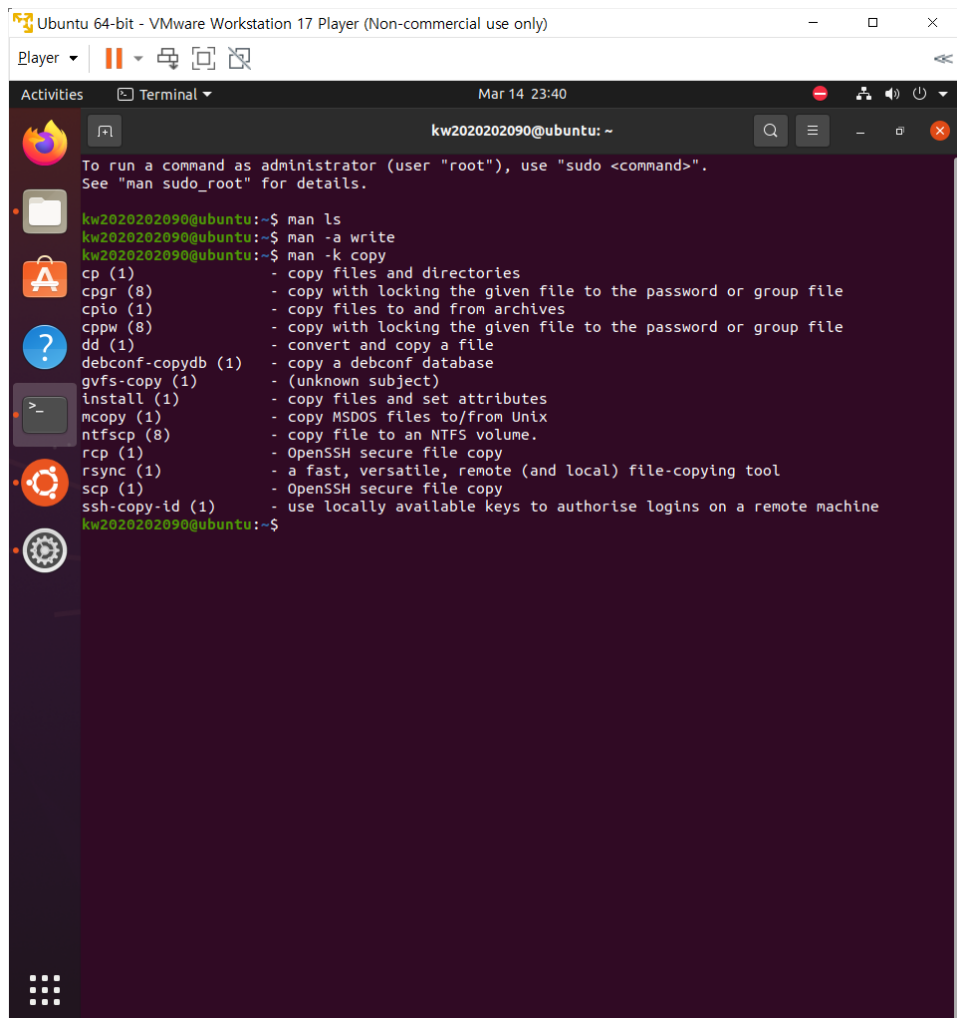
  --color[=WHEN]
      colorize the output; WHEN can be 'always' (default if omitted),
      'auto', or 'never'; more info below

  -d, --directory
      list directories themselves, not their contents

Manual page ls(1) line 1 (press h for help or q to quit)
```

ls 명령어에 대한 manual이 출력된다.

■ man -k copy



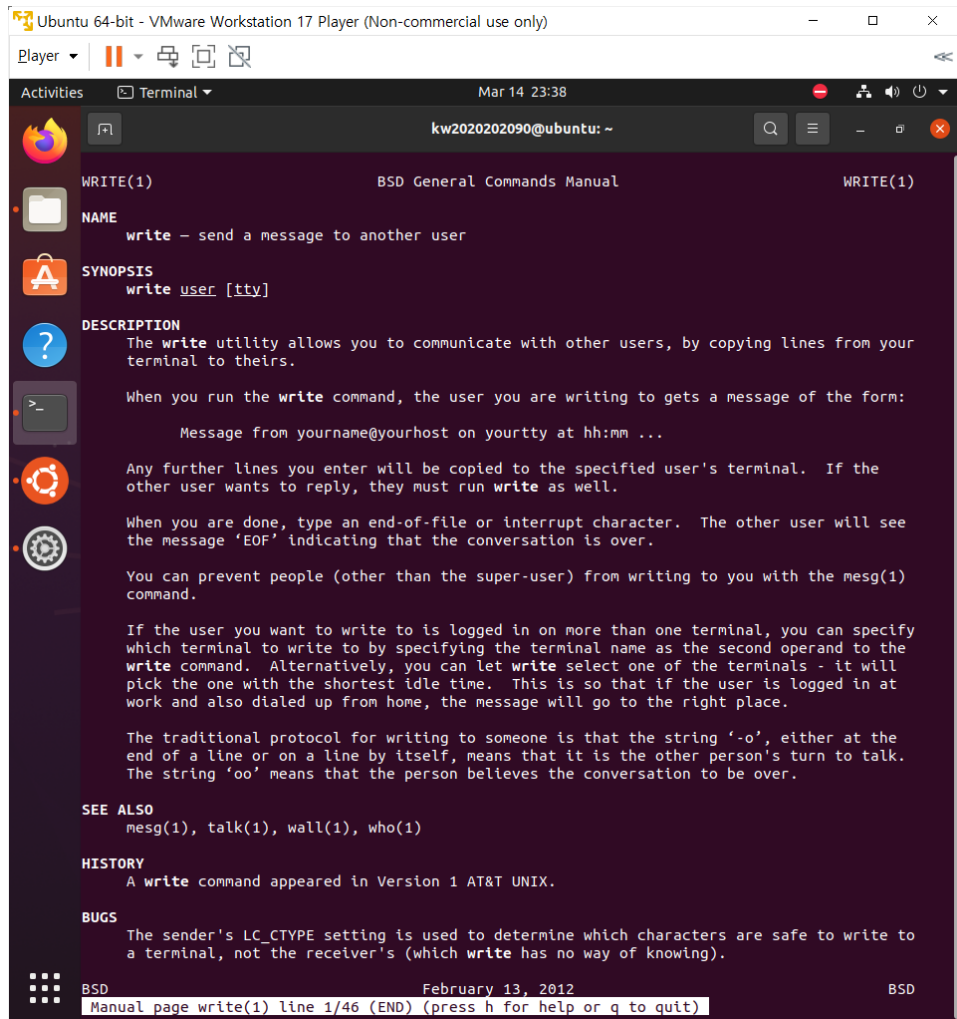
```
Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)
Player
Activities Terminal Mar 14 23:40 kw2020202090@ubuntu: ~

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

kw2020202090@ubuntu:~$ man ls
kw2020202090@ubuntu:~$ man -a write
kw2020202090@ubuntu:~$ man -k copy
cp (1) - copy files and directories
cpgr (8) - copy with locking the given file to the password or group file
cpio (1) - copy files to and from archives
cppw (8) - copy with locking the given file to the password or group file
dd (1) - convert and copy a file
debconf-copydb (1) - copy a debconf database
gvfs-copy (1) - (unknown subject)
install (1) - copy files and set attributes
mcopy (1) - copy MSDOS files to/from Unix
ntfscp (8) - copy file to an NTFS volume.
rcp (1) - OpenSSH secure file copy
rsync (1) - a fast, versatile, remote (and local) file-copying tool
scp (1) - OpenSSH secure file copy
ssh-copy-id (1) - use locally available keys to authorise logins on a remote machine
kw2020202090@ubuntu:~$
```

copy 명령어에 대한 keyword search 결과를 출력한다.

■ man -a write



The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal is displaying the manual page for the 'write' command. The window has a dark theme with a sidebar on the left containing icons for various applications. The terminal output is as follows:

```
WRITE(1)                                BSD General Commands Manual                                WRITE(1)

NAME
    write - send a message to another user

SYNOPSIS
    write user [tty]

DESCRIPTION
    The write utility allows you to communicate with other users, by copying lines from your terminal to theirs.

    When you run the write command, the user you are writing to gets a message of the form:

        Message from yourname@yourhost on yourtty at hh:mm ...

    Any further lines you enter will be copied to the specified user's terminal. If the other user wants to reply, they must run write as well.

    When you are done, type an end-of-file or interrupt character. The other user will see the message 'EOF' indicating that the conversation is over.

    You can prevent people (other than the super-user) from writing to you with the mesg(1) command.

    If the user you want to write to is logged in on more than one terminal, you can specify which terminal to write to by specifying the terminal name as the second operand to the write command. Alternatively, you can let write select one of the terminals - it will pick the one with the shortest idle time. This is so that if the user is logged in at work and also dialed up from home, the message will go to the right place.

    The traditional protocol for writing to someone is that the string '-o', either at the end of a line or on a line by itself, means that it is the other person's turn to talk. The string 'oo' means that the person believes the conversation to be over.

SEE ALSO
    mesg(1), talk(1), wall(1), who(1)

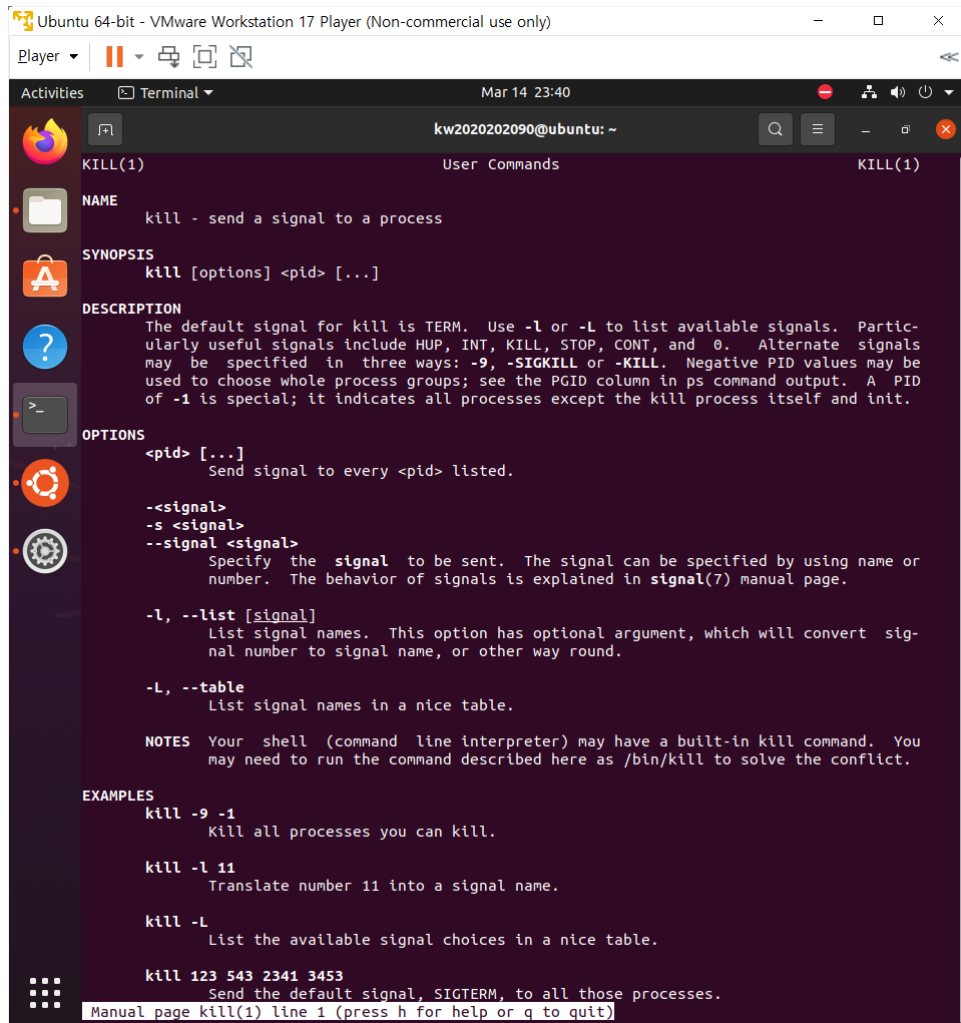
HISTORY
    A write command appeared in Version 1 AT&T UNIX.

BUGS
    The sender's LC_CTYPE setting is used to determine which characters are safe to write to a terminal, not the receiver's (which write has no way of knowing).

BSD                                     February 13, 2012                                     BSD
Manual page write(1) line 1/46 (END) (press h for help or q to quit)
```

write 명령어에 대한 모든 manual 정보를 출력한다.

■ man kill



Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)

Player ▾

Activities Terminal ▾ Mar 14 23:40

kw2020202090@ubuntu: ~

KILL(1) User Commands KILL(1)

NAME
kill - send a signal to a process

SYNOPSIS
kill [options] <pid> [...]

DESCRIPTION
The default signal for kill is TERM. Use -l or -L to list available signals. Particularly useful signals include HUP, INT, KILL, STOP, CONT, and 0. Alternate signals may be specified in three ways: -9, -SIGKILL or -KILL. Negative PID values may be used to choose whole process groups; see the PGID column in ps command output. A PID of -1 is special; it indicates all processes except the kill process itself and init.

OPTIONS
<pid> [...]
Send signal to every <pid> listed.

-<signal>
-s <signal>
--signal <signal>
Specify the signal to be sent. The signal can be specified by using name or number. The behavior of signals is explained in **signal(7)** manual page.

-l, --list [signal]
List signal names. This option has optional argument, which will convert signal number to signal name, or other way round.

-L, --table
List signal names in a nice table.

NOTES Your shell (command line interpreter) may have a built-in kill command. You may need to run the command described here as /bin/kill to solve the conflict.

EXAMPLES
kill -9 -1
Kill all processes you can kill.

kill -l 11
Translate number 11 into a signal name.

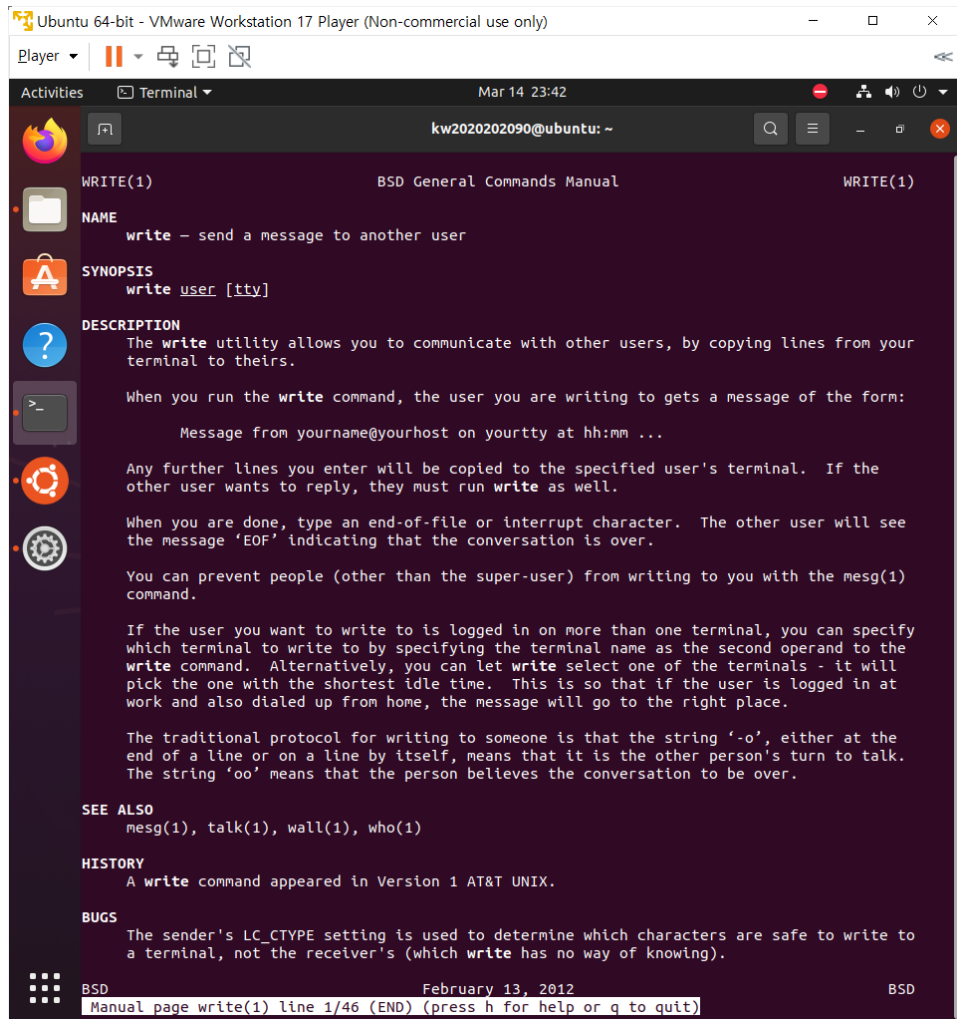
kill -L
List the available signal choices in a nice table.

kill 123 543 2341 3453
Send the default signal, SIGTERM, to all those processes.

Manual page kill(1) line 1 (press h for help or q to quit)

kill 명령어에 대한 manual이 출력된다.

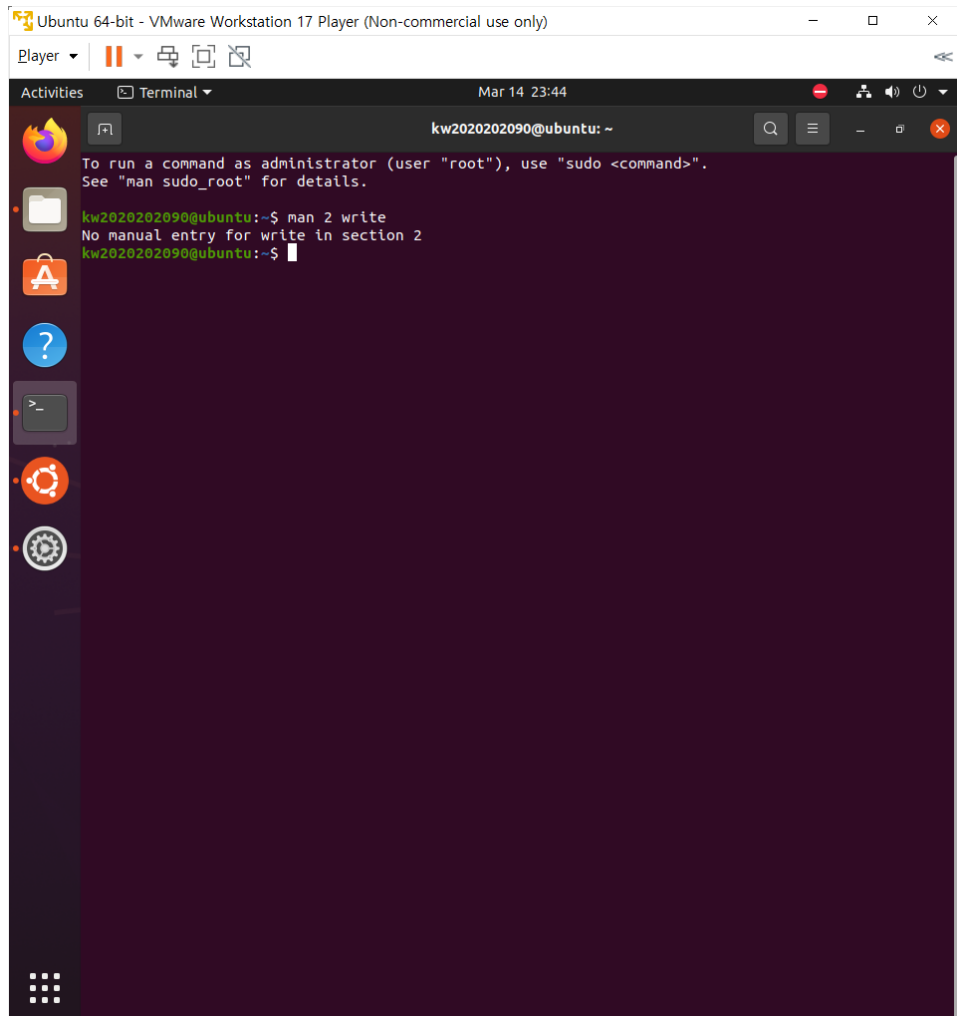
■ man 1 write



```
Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)
Player
Activities Terminal Mar 14 23:42 kw2020202090@ubuntu: ~
WRITE(1) BSD General Commands Manual WRITE(1)
NAME
  write - send a message to another user
SYNOPSIS
  write user [tty]
DESCRIPTION
  The write utility allows you to communicate with other users, by copying lines from your terminal to theirs.
  When you run the write command, the user you are writing to gets a message of the form:
    Message from yourname@yourhost on yourtty at hh:mm ...
  Any further lines you enter will be copied to the specified user's terminal. If the other user wants to reply, they must run write as well.
  When you are done, type an end-of-file or interrupt character. The other user will see the message 'EOF' indicating that the conversation is over.
  You can prevent people (other than the super-user) from writing to you with the mesg(1) command.
  If the user you want to write to is logged in on more than one terminal, you can specify which terminal to write to by specifying the terminal name as the second operand to the write command. Alternatively, you can let write select one of the terminals - it will pick the one with the shortest idle time. This is so that if the user is logged in at work and also dialed up from home, the message will go to the right place.
  The traditional protocol for writing to someone is that the string '-o', either at the end of a line or on a line by itself, means that it is the other person's turn to talk. The string 'oo' means that the person believes the conversation to be over.
SEE ALSO
  mesg(1), talk(1), wall(1), who(1)
HISTORY
  A write command appeared in Version 1 AT&T UNIX.
BUGS
  The sender's LC_CTYPE setting is used to determine which characters are safe to write to a terminal, not the receiver's (which write has no way of knowing).
BSD February 13, 2012
Manual page write(1) line 1/46 (END) (press h for help or q to quit)
```

write 명령어에 대한 general commands manual을 출력한다.

■ man 2 write

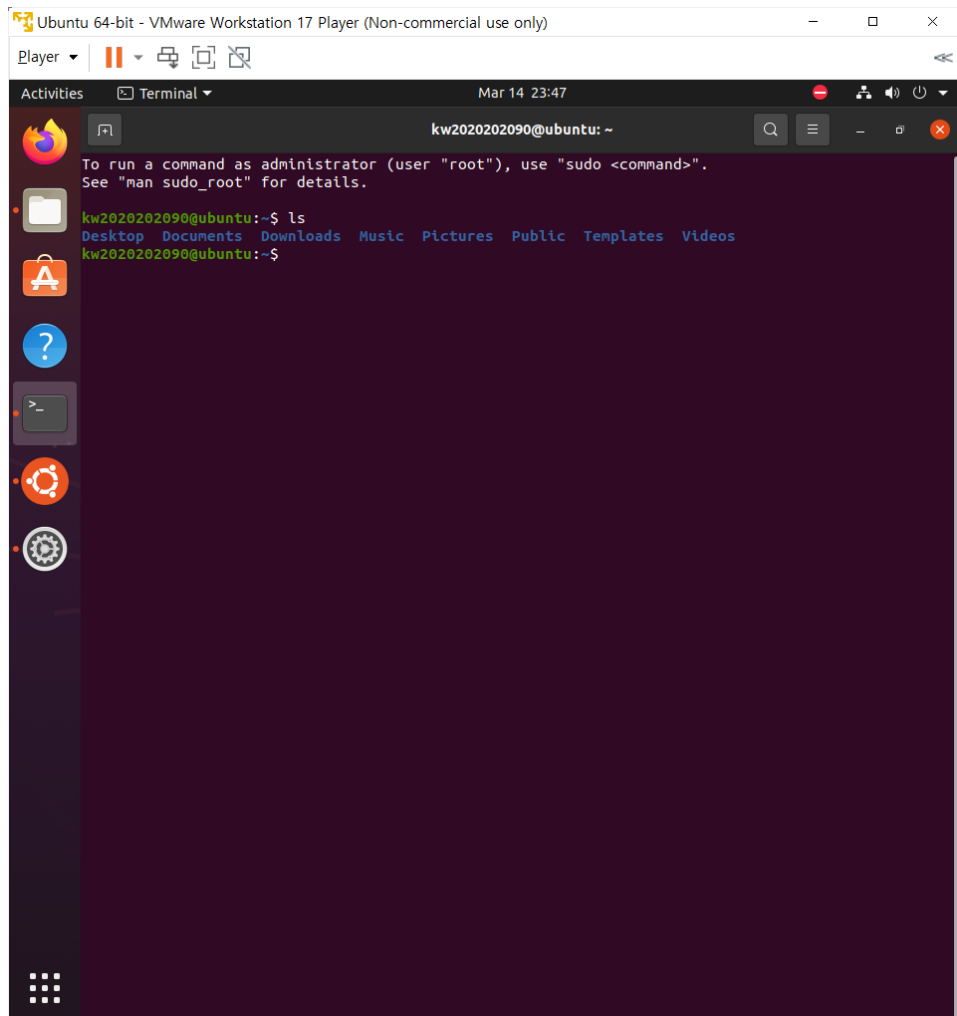


```
Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)
Player
Activities Terminal Mar 14 23:44 kw2020202090@ubuntu: ~
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
kw2020202090@ubuntu:~$ man 2 write
No manual entry for write in section 2
kw2020202090@ubuntu:~$
```

write 명령어에 대한 system calls manual을 출력하는 명령어이다.
write 명령어는 system call에 대한 manual이 존재하지 않는다고 출력
됐다.

- ls

■ ls

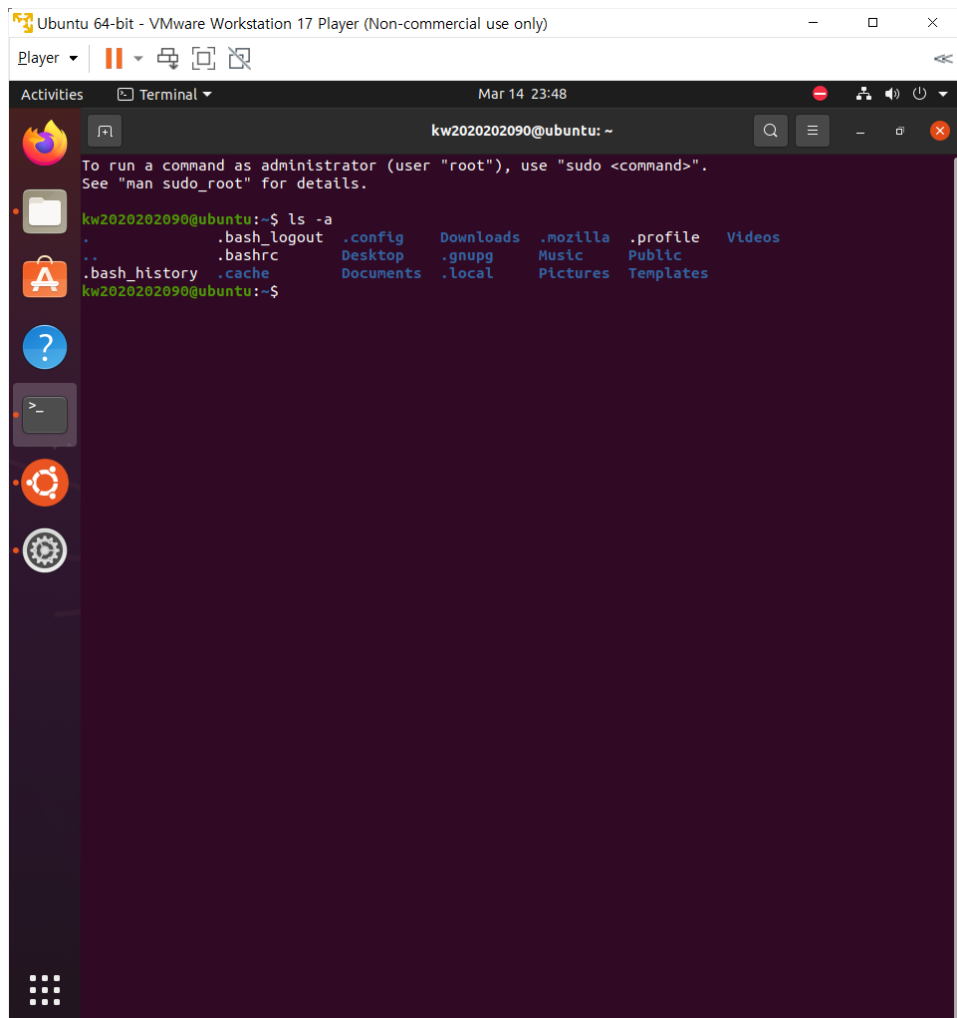


The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal prompt is "kw2020202090@ubuntu: ~". The user has entered the command "ls", and the output is displayed as follows:

```
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
kw2020202090@ubuntu:~$ ls  
Desktop Documents Downloads Music Pictures Public Templates Videos  
kw2020202090@ubuntu:~$
```

현재 디렉토리의 모든 콘텐츠 목록을 출력한다.

■ ls -a

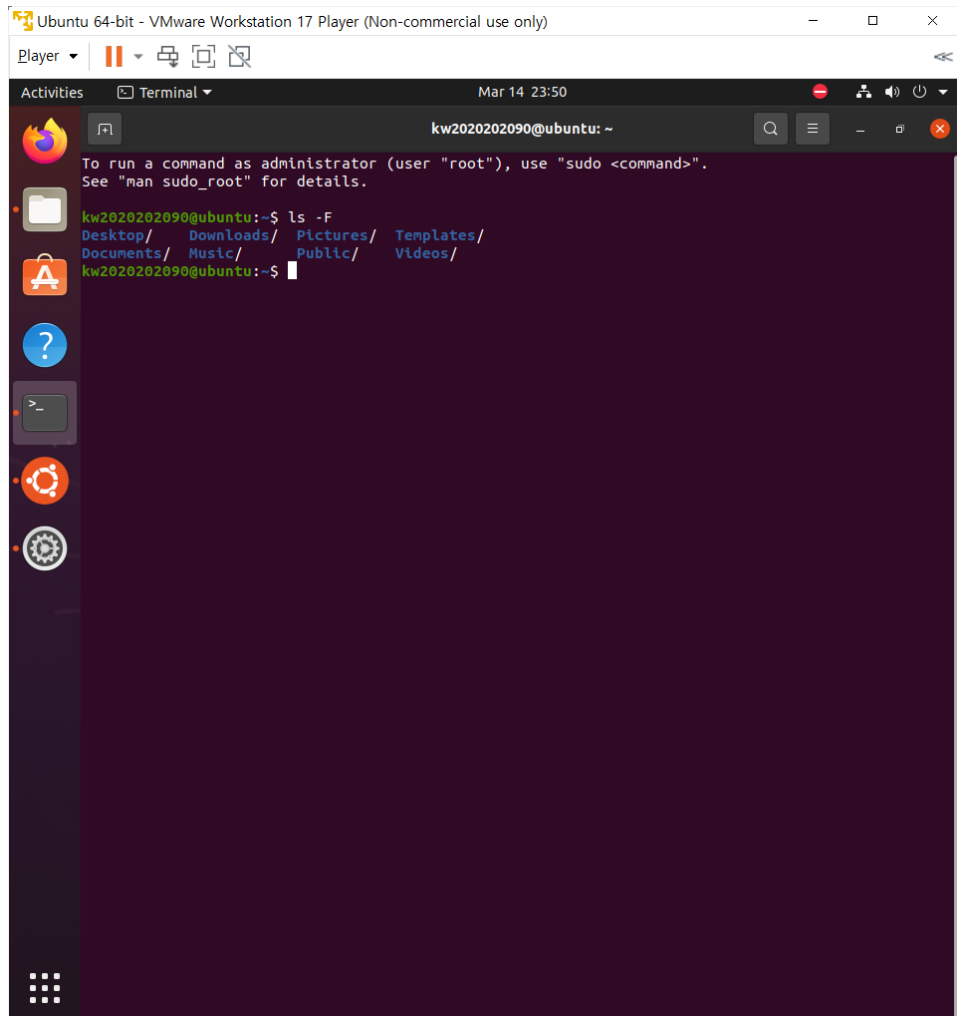


The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal displays the following text:

```
kw2020202090@ubuntu: ~  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
kw2020202090@ubuntu:~$ ls -a  
..      .bash_logout .config  Downloads .mozilla  .profile  Videos  
..      .bashrc      Desktop  .gnupg    Music     Public  
..      .bash_history .cache   Documents .local    Pictures  Templates  
kw2020202090@ubuntu:~$
```

숨김 파일을 포함한 모든 파일을 출력한다.

■ ls -f

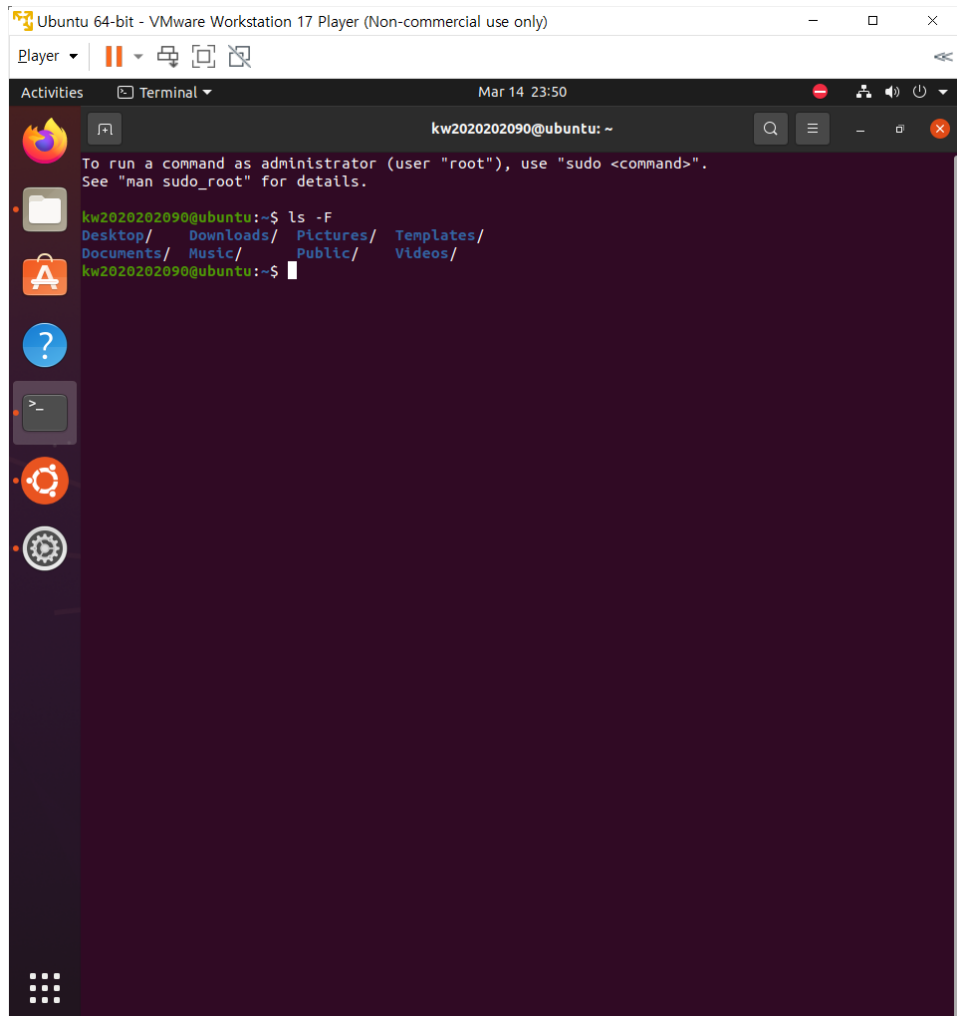


The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal prompt is "kw2020202090@ubuntu: ~". The command "ls -f" has been executed, resulting in the following output:

```
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
kw2020202090@ubuntu:~$ ls -f  
Desktop/  Downloads/  Pictures/  Templates/  
Documents/ Music/      Public/    Videos/  
kw2020202090@ubuntu:~$
```

현재 디렉토리의 모든 파일과 종류를 표시한다.

■ ls -l

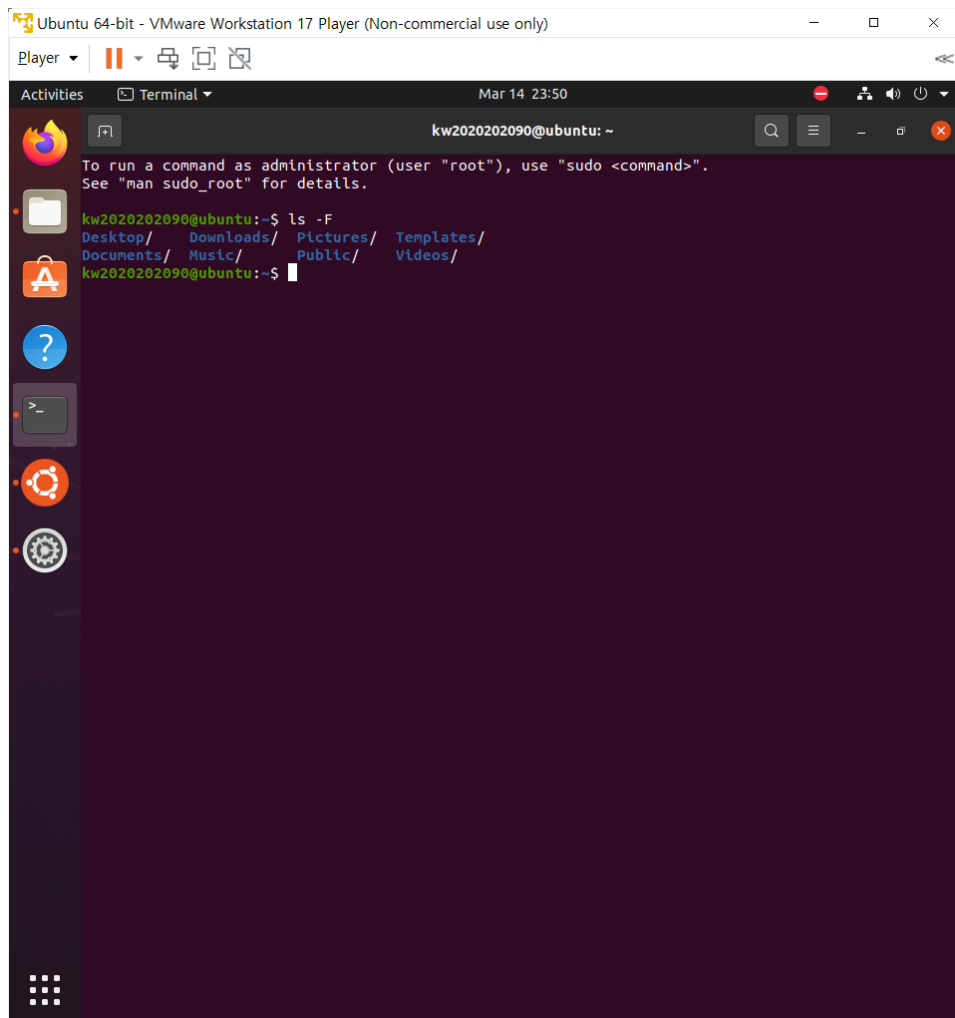


The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal prompt is "kw2020202090@ubuntu: ~". The command "ls -l" has been executed, resulting in the following output:

```
kw2020202090@ubuntu:~$ ls -l
Desktop/  Downloads/  Pictures/  Templates/
Documents/ Music/      Public/    Videos/
kw2020202090@ubuntu:~$
```

현재 디렉토리의 모든 파일을 자세히 출력한다.

- pwd



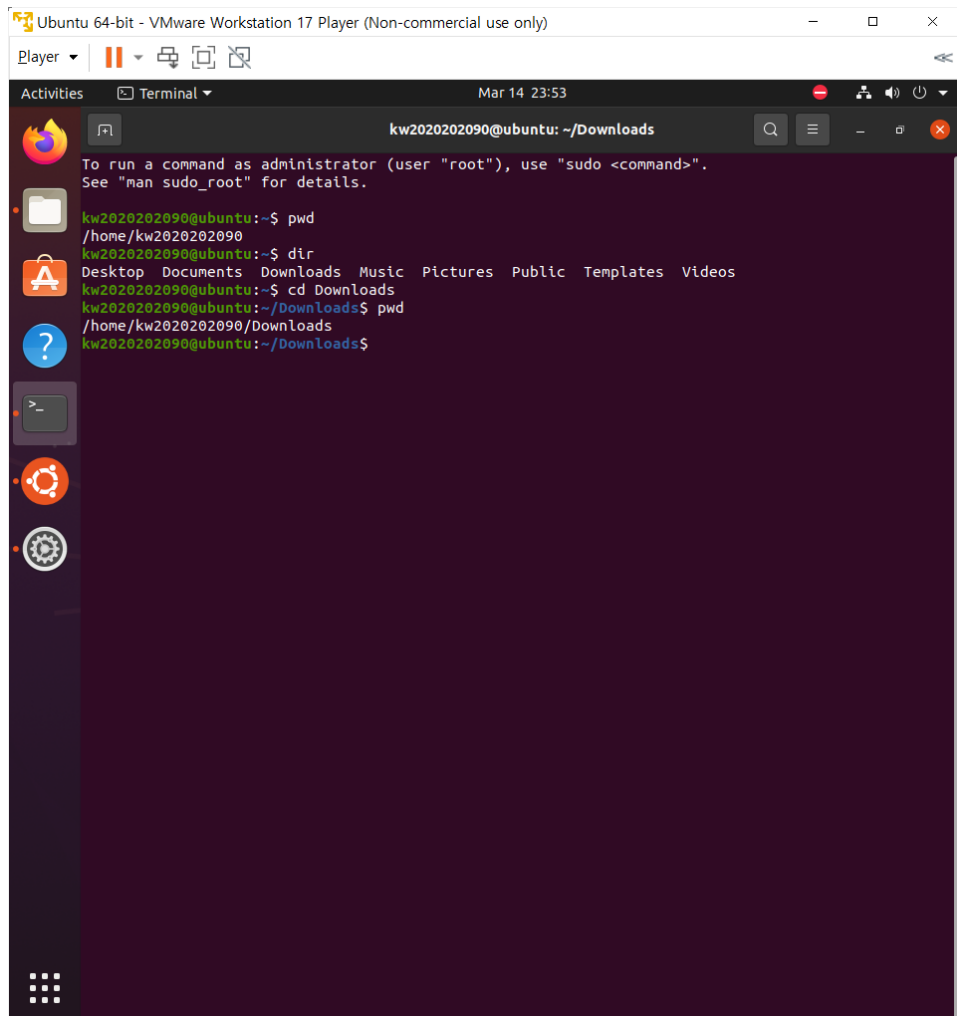
The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal prompt is "kw2020202090@ubuntu: ~". The output of the command "ls -F" is displayed as follows:

```
kw2020202090@ubuntu:~$ ls -F
Desktop/  Downloads/  Pictures/  Templates/
Documents/ Music/      Public/    Videos/
kw2020202090@ubuntu:~$
```

현재 작업 디렉토리를 출력한다.

- cd

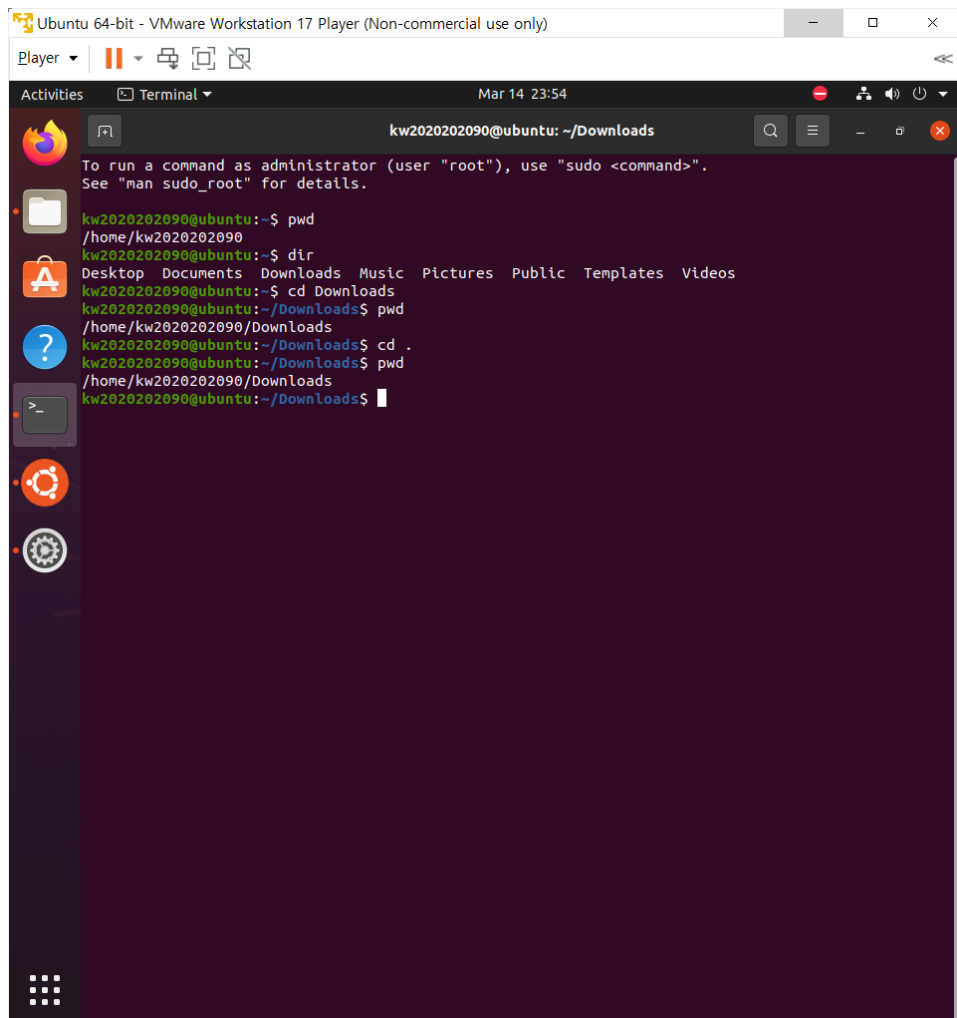
■ cd (dir)



```
Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)
Player
Activities Terminal Mar 14 23:53
kw2020202090@ubuntu: ~/Downloads
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
kw2020202090@ubuntu:~$ pwd
/home/kw2020202090
kw2020202090@ubuntu:~$ dir
Desktop Documents Downloads Music Pictures Public Templates Videos
kw2020202090@ubuntu:~$ cd Downloads
kw2020202090@ubuntu:~/Downloads$ pwd
/home/kw2020202090/Downloads
kw2020202090@ubuntu:~/Downloads$
```

Downloads 디렉토리로 이동하였다.

■ cd .

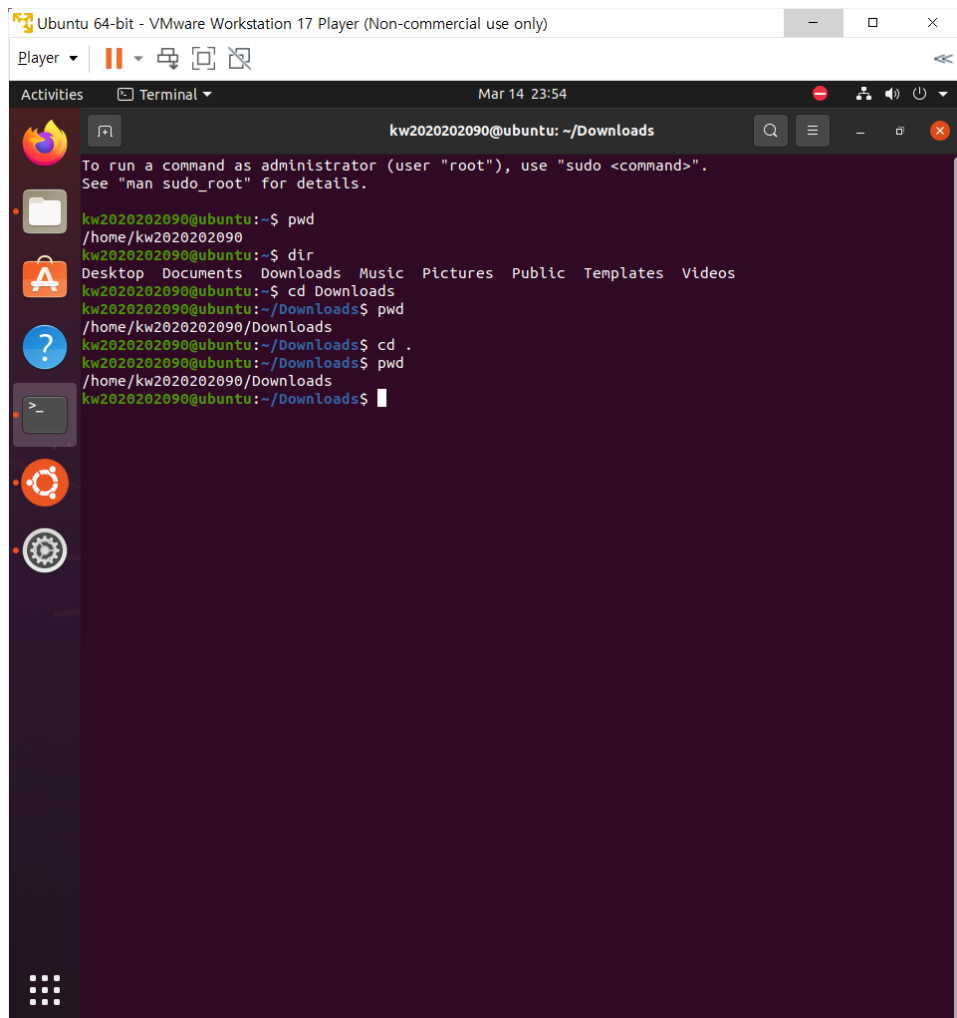


The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal is running on a user named "kw2020202090@ubuntu" in the directory "~/Downloads". The terminal output shows the following commands and their results:

```
kw2020202090@ubuntu:~$ pwd
/home/kw2020202090
kw2020202090@ubuntu:~$ dir
Desktop Documents Downloads Music Pictures Public Templates Videos
kw2020202090@ubuntu:~$ cd Downloads
kw2020202090@ubuntu:~/Downloads$ pwd
/home/kw2020202090/Downloads
kw2020202090@ubuntu:~/Downloads$ cd .
kw2020202090@ubuntu:~/Downloads$ pwd
/home/kw2020202090/Downloads
kw2020202090@ubuntu:~/Downloads$
```

현재 디렉토리로 이동하였다.

■ cd ..

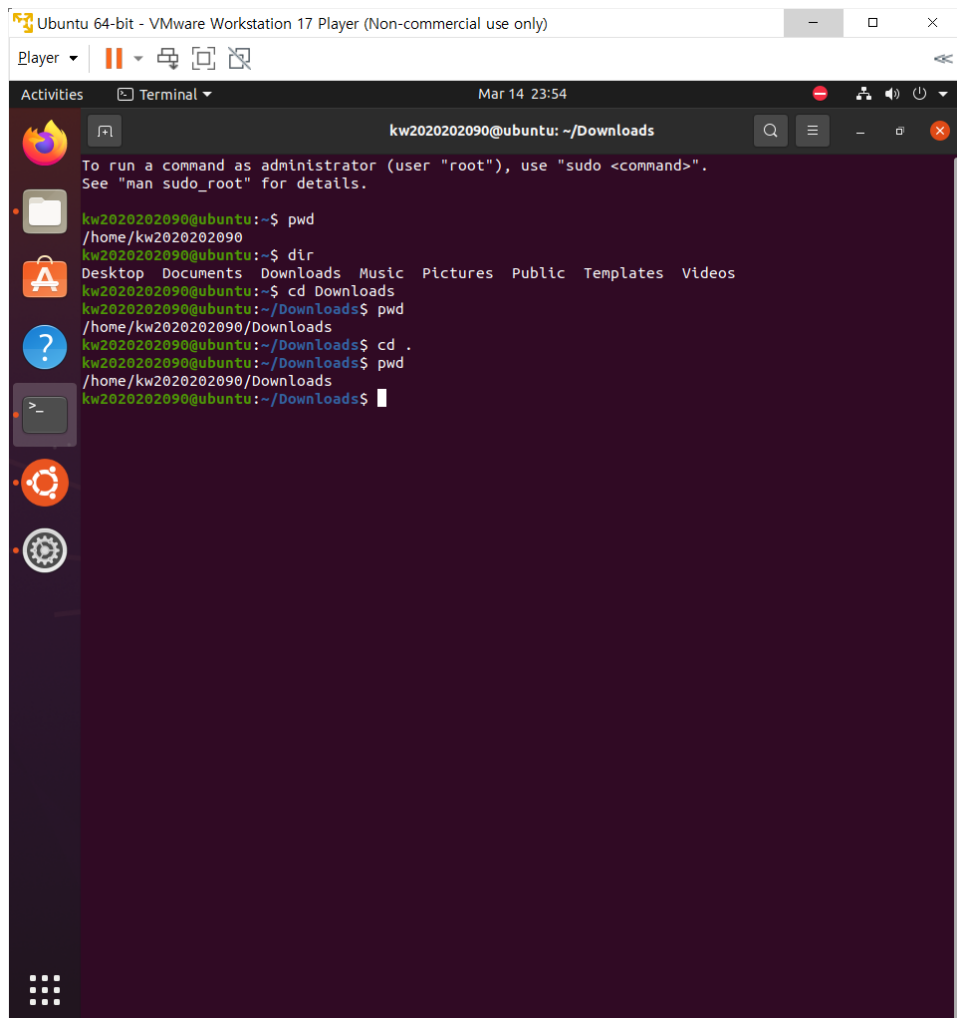


The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal output is as follows:

```
kw2020202090@ubuntu: ~/Downloads
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
kw2020202090@ubuntu:~$ pwd
/home/kw2020202090
kw2020202090@ubuntu:~$ dir
Desktop Documents Downloads Music Pictures Public Templates Videos
kw2020202090@ubuntu:~$ cd Downloads
kw2020202090@ubuntu:~/Downloads$ pwd
/home/kw2020202090/Downloads
kw2020202090@ubuntu:~/Downloads$ cd .
kw2020202090@ubuntu:~/Downloads$ pwd
/home/kw2020202090/Downloads
kw2020202090@ubuntu:~/Downloads$
```

Downloads의 상위 디렉토리인 home 디렉토리로 이동하였다.

■ cd ~



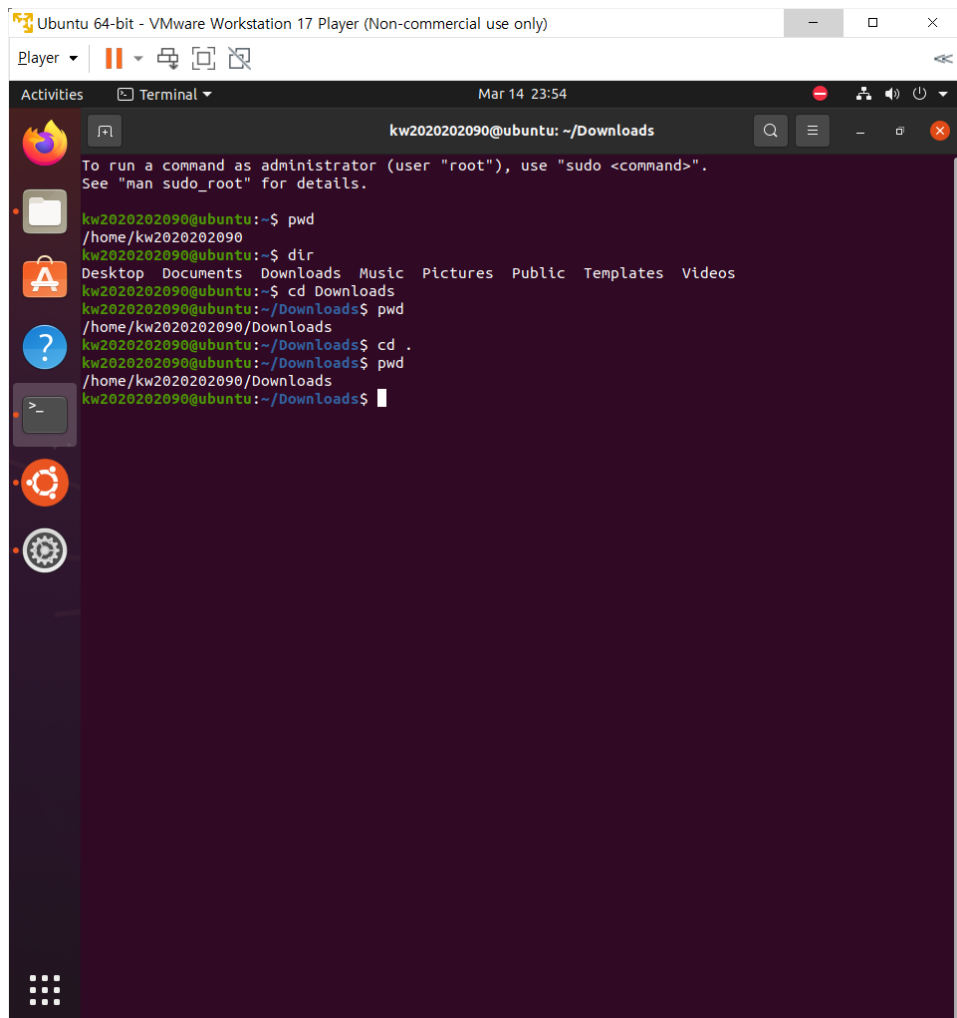
The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal output is as follows:

```
kw2020202090@ubuntu: ~/Downloads
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
kw2020202090@ubuntu:~$ pwd
/home/kw2020202090
kw2020202090@ubuntu:~$ dir
Desktop Documents Downloads Music Pictures Public Templates Videos
kw2020202090@ubuntu:~$ cd Downloads
kw2020202090@ubuntu:~/Downloads$ pwd
/home/kw2020202090/Downloads
kw2020202090@ubuntu:~/Downloads$ cd .
kw2020202090@ubuntu:~/Downloads$ pwd
/home/kw2020202090/Downloads
kw2020202090@ubuntu:~/Downloads$
```

The terminal window also shows a sidebar with application icons and a top bar with system status indicators.

Documents 디렉토리에서 home 디렉토리로 바로 이동하였다.

■ cd -



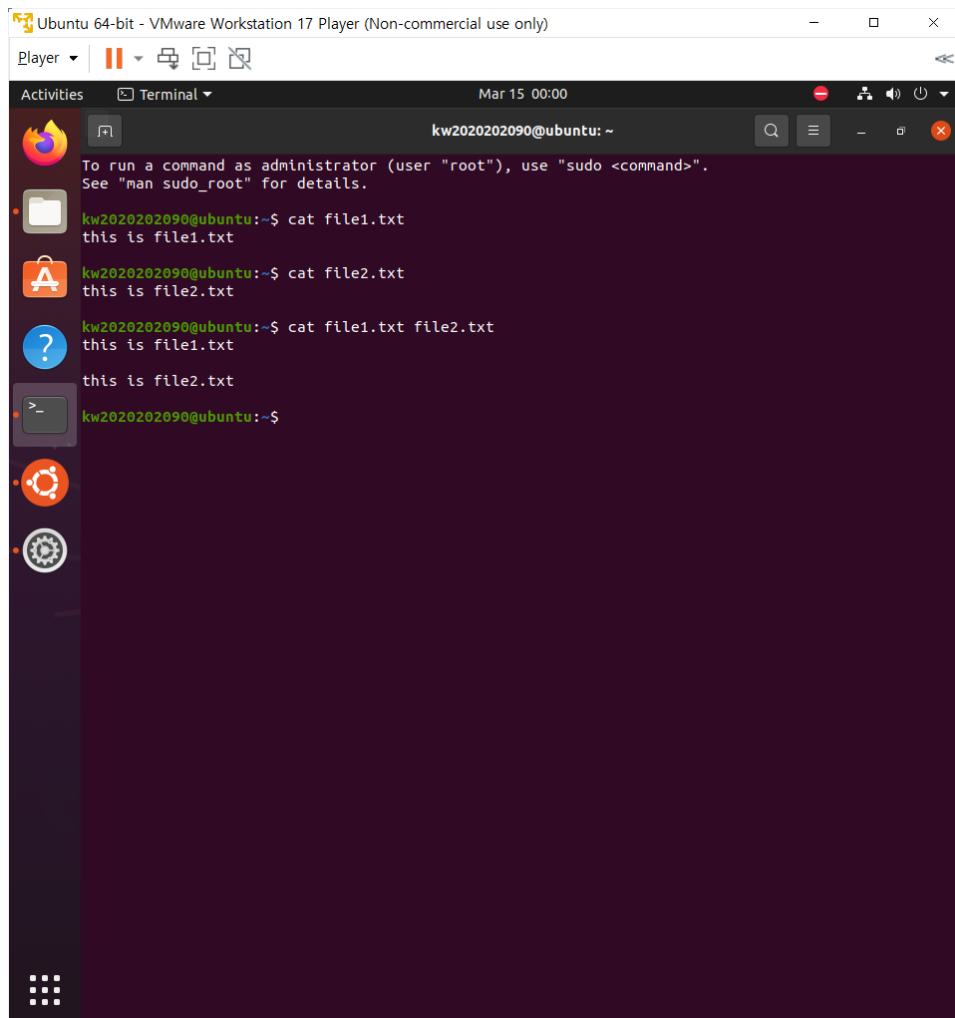
The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal output is as follows:

```
kw2020202090@ubuntu: ~/$ pwd
/home/kw2020202090
kw2020202090@ubuntu:~$ dir
Desktop Documents Downloads Music Pictures Public Templates Videos
kw2020202090@ubuntu:~$ cd Downloads
kw2020202090@ubuntu:~/Downloads$ pwd
/home/kw2020202090/Downloads
kw2020202090@ubuntu:~/Downloads$ cd .
kw2020202090@ubuntu:~/Downloads$ pwd
/home/kw2020202090/Downloads
kw2020202090@ubuntu:~/Downloads$
```

The terminal window includes a sidebar with application icons (Firefox, Files, App Store, Dash, Question mark, Terminal, and Settings) and a top bar with system status icons (network, volume, power) and the date/time "Mar 14 23:54".

이전 디렉토리인 Documents 디렉토리로 다시 돌아갔다.

- cat



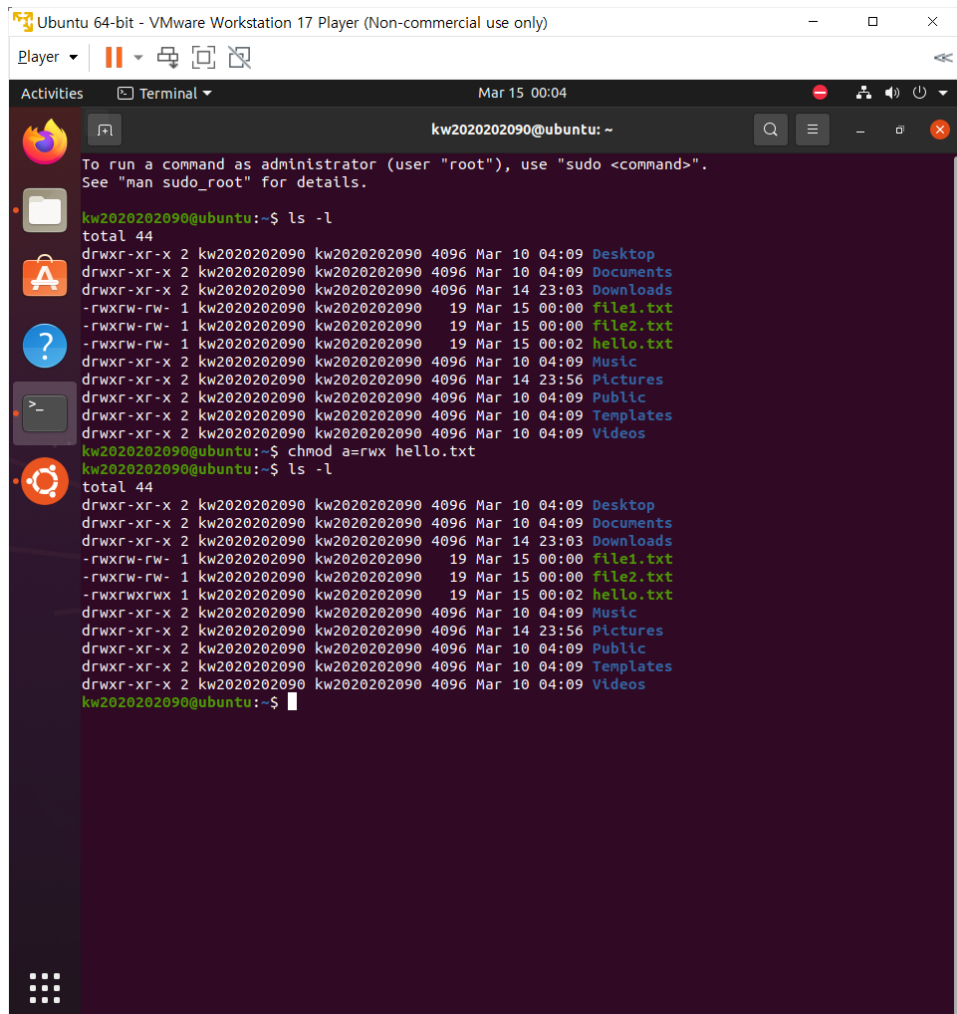
The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal prompt is "kw2020202090@ubuntu: ~". The terminal output shows the following commands and their results:

```
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
kw2020202090@ubuntu:~$ cat file1.txt  
this is file1.txt  
  
kw2020202090@ubuntu:~$ cat file2.txt  
this is file2.txt  
  
kw2020202090@ubuntu:~$ cat file1.txt file2.txt  
this is file1.txt  
this is file2.txt  
  
kw2020202090@ubuntu:~$
```

파일의 내용을 합쳐서 출력한다.

- chmod

■ chmod a=rwx (file)

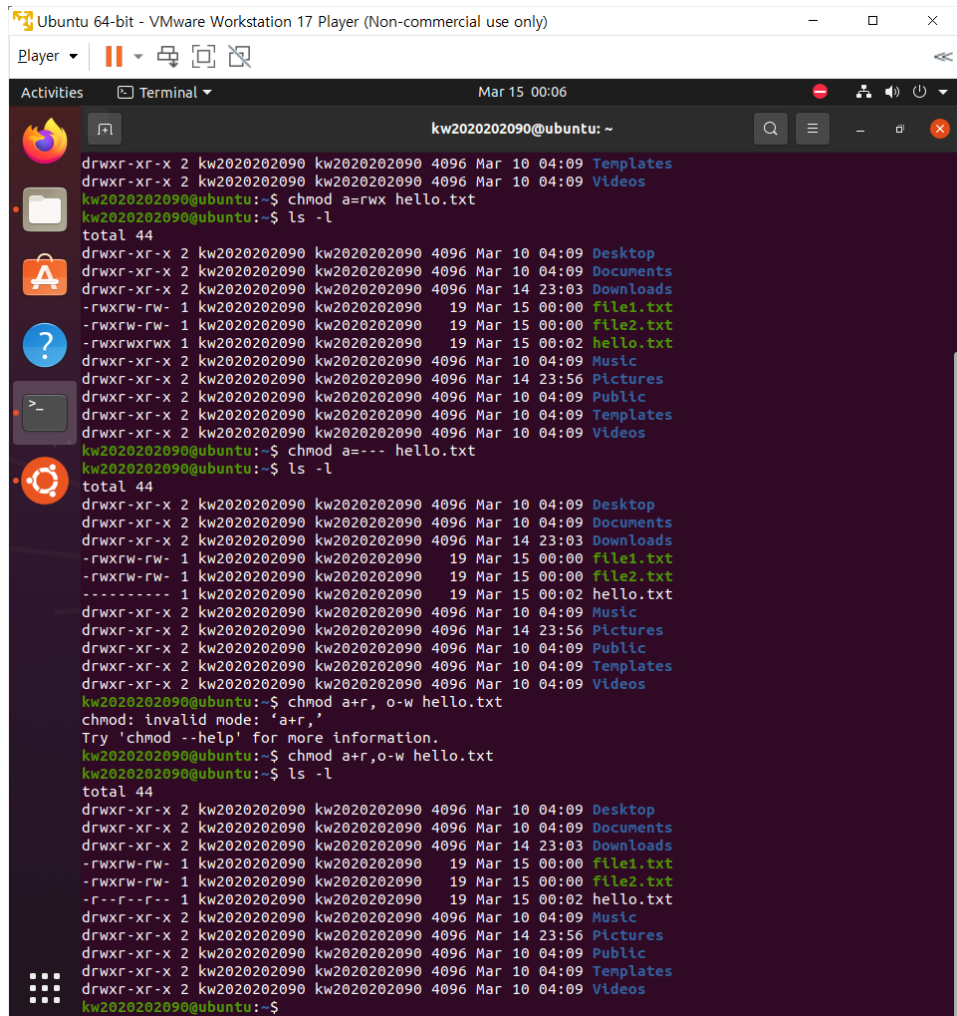


The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal output is as follows:

```
kw2020202090@ubuntu: ~  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
kw2020202090@ubuntu:~$ ls -l  
total 44  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads  
-rw-rw-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file1.txt  
-rw-rw-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file2.txt  
-rw-rw-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:56 Pictures  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos  
kw2020202090@ubuntu:~$ chmod a=rwx hello.txt  
kw2020202090@ubuntu:~$ ls -l  
total 44  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads  
-rw-rw-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file1.txt  
-rw-rw-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file2.txt  
-rwxrwxrwx 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:56 Pictures  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos  
kw2020202090@ubuntu:~$
```

hello.txt 파일의 권한이 모두에게 읽기, 쓰기, 실행이 가능한 상태로 바뀌었다.

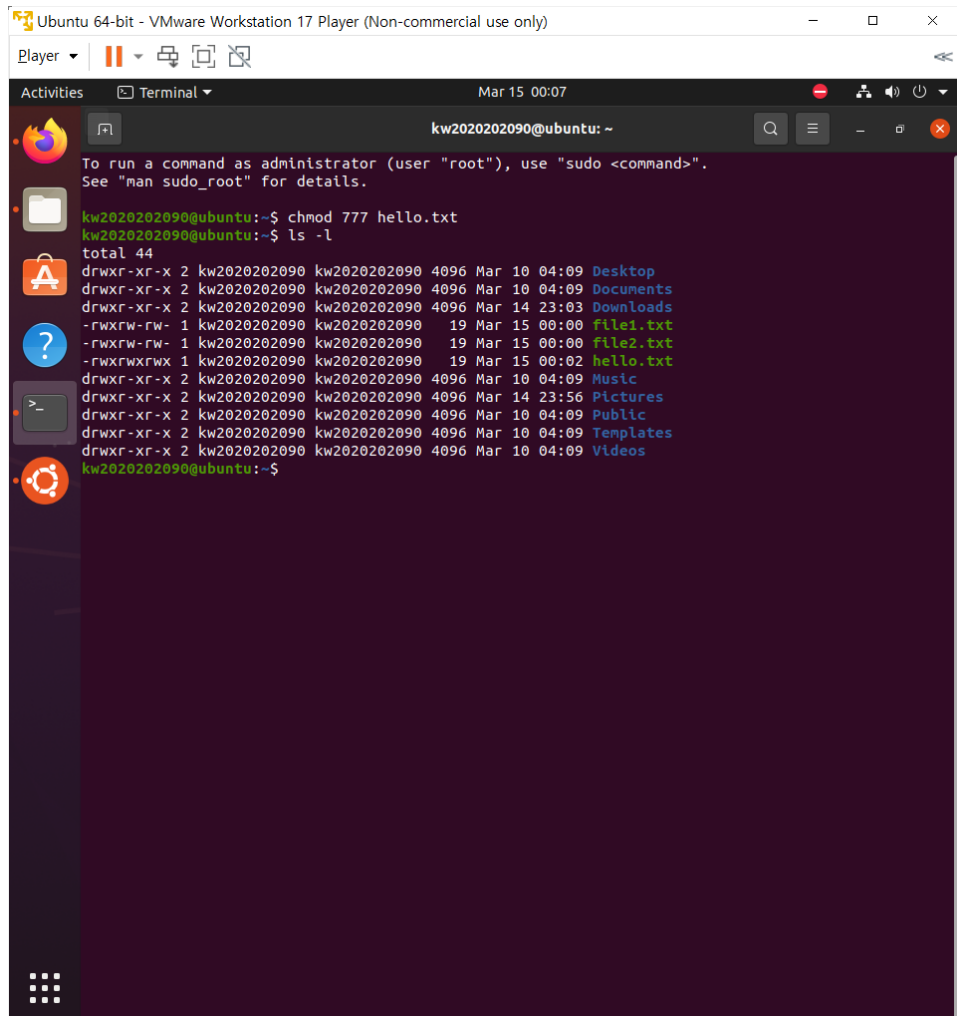
■ chmod a+r,o-w (file)



```
Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)
Player
Activities Terminal Mar 15 00:06
kw2020202090@ubuntu: ~
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos
kw2020202090@ubuntu:~$ chmod a=rwx hello.txt
kw2020202090@ubuntu:~$ ls -l
total 44
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads
-rwxrwxrwx- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file1.txt
-rwxrwxrwx- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file2.txt
-rwxrwxrwx- 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:56 Pictures
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos
kw2020202090@ubuntu:~$ chmod a=--- hello.txt
kw2020202090@ubuntu:~$ ls -l
total 44
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads
-rwxrwxrwx- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file1.txt
-rwxrwxrwx- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file2.txt
----- 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:56 Pictures
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos
kw2020202090@ubuntu:~$ chmod a+r, o-w hello.txt
chmod: invalid mode: 'a+r,'
Try 'chmod --help' for more information.
kw2020202090@ubuntu:~$ chmod a+r,o-w hello.txt
kw2020202090@ubuntu:~$ ls -l
total 44
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads
-rwxrwxrwx- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file1.txt
-rwxrwxrwx- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file2.txt
-r--r--r-- 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:56 Pictures
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos
kw2020202090@ubuntu:~$
```

hello.txt 파일의 권한이 모두에게 읽기 권한이 주어지고 others는 실행 권한이 삭제됐다.

■ chmod 777 (file)

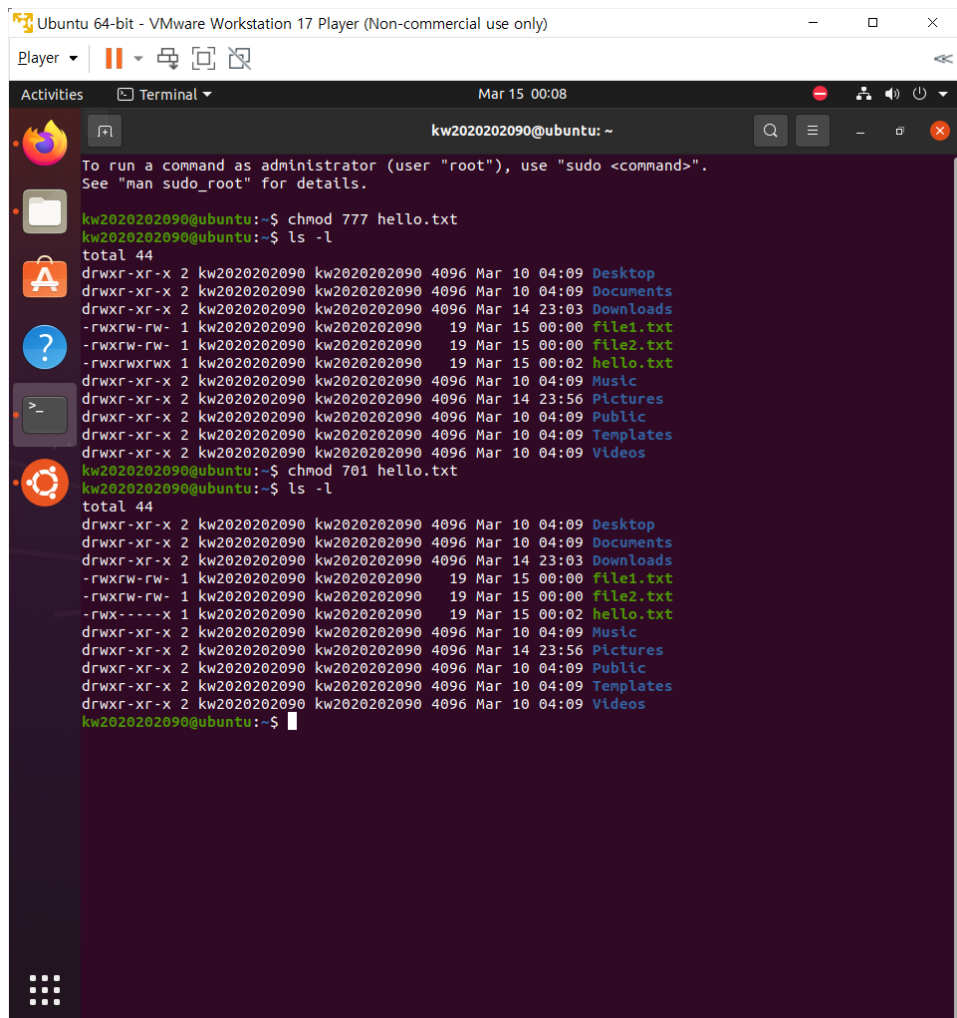


The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal output is as follows:

```
kw2020202090@ubuntu: ~  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
kw2020202090@ubuntu:~$ chmod 777 hello.txt  
kw2020202090@ubuntu:~$ ls -l  
total 44  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads  
-rwxrwxrwx 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file1.txt  
-rwxrwxrwx 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file2.txt  
-rwxrwxrwx 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:56 Pictures  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos  
kw2020202090@ubuntu:~$
```

hello.txt 파일에 대해 모든 대상에게 모든 권한이 부여됐다.

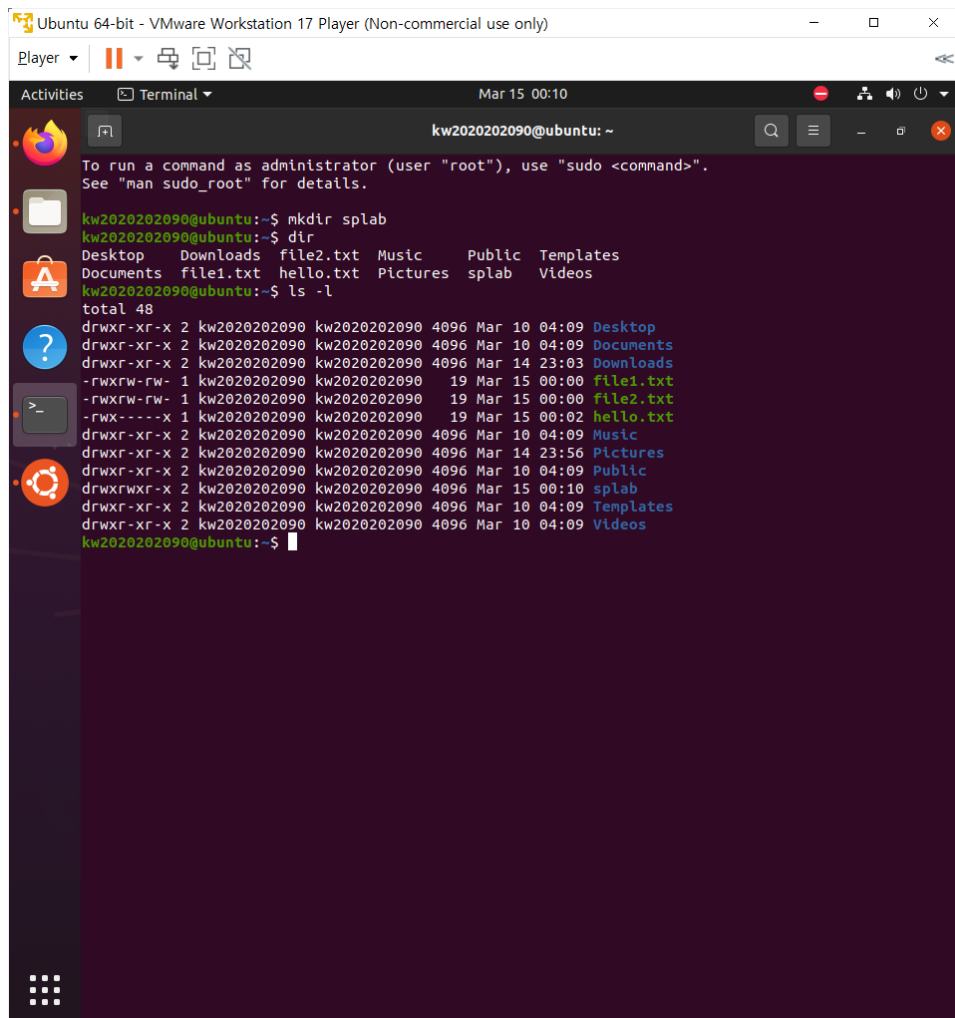
■ chmod 701 (file)



```
Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)
Player
Activities Terminal Mar 15 00:08
kw2020202090@ubuntu: ~
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
kw2020202090@ubuntu:~$ chmod 777 hello.txt
kw2020202090@ubuntu:~$ ls -l
total 44
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads
-rwxrwx-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file1.txt
-rwxrwx-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file2.txt
-rwxrwxrwx 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:56 Pictures
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos
kw2020202090@ubuntu:~$ chmod 701 hello.txt
kw2020202090@ubuntu:~$ ls -l
total 44
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads
-rwxrwx-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file1.txt
-rwxrwx-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file2.txt
-rwx----x 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:56 Pictures
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos
kw2020202090@ubuntu:~$
```

hello.txt 파일에 대해 소유자에겐 모든 권한이, others는 실행만 가능하도록 권한이 변경되었다.

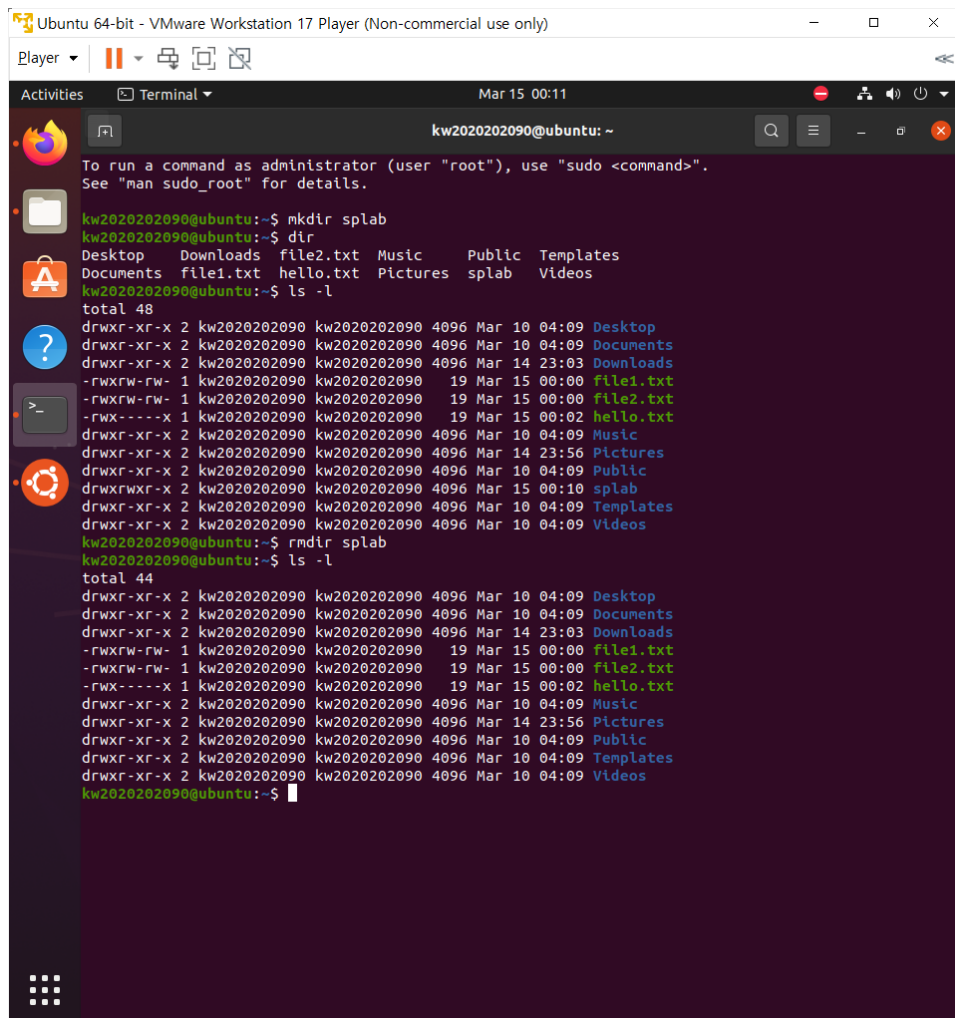
- mkdir (directory)



```
Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)
Player
Activities Terminal Mar 15 00:10
kw2020202090@ubuntu: ~
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
kw2020202090@ubuntu:~$ mkdir splab
kw2020202090@ubuntu:~$ dir
Desktop Downloads file2.txt Music Public Templates
Documents file1.txt hello.txt Pictures splab Videos
kw2020202090@ubuntu:~$ ls -l
total 48
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads
-rwxrw-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file1.txt
-rwxrw-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file2.txt
-rwx----x 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:56 Pictures
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public
drwxrwxr-x 2 kw2020202090 kw2020202090 4096 Mar 15 00:10 splab
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos
kw2020202090@ubuntu:~$
```

home 디렉토리에 splab이라는 디렉토리가 생성되었다.

- rmdir (directory)

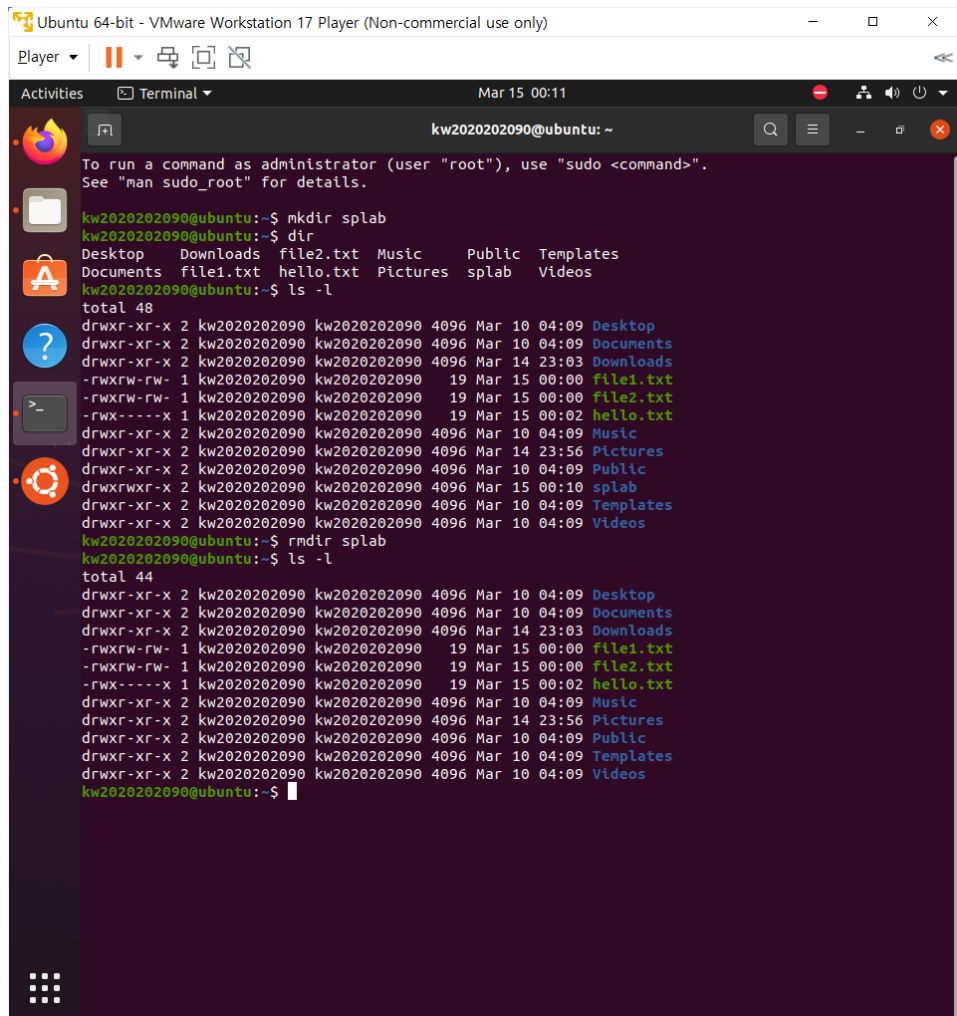


```
Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)
Player
Activities Terminal Mar 15 00:11
kw2020202090@ubuntu: ~
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
kw2020202090@ubuntu:~$ mkdir splab
kw2020202090@ubuntu:~$ dir
Desktop Downloads file2.txt Music Public Templates
Documents file1.txt hello.txt Pictures splab Videos
kw2020202090@ubuntu:~$ ls -l
total 48
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads
-rwxrw-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file1.txt
-rwxrw-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file2.txt
-rwx----x 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:56 Pictures
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public
drwxrwxr-x 2 kw2020202090 kw2020202090 4096 Mar 15 00:10 splab
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos
kw2020202090@ubuntu:~$ rmdir splab
kw2020202090@ubuntu:~$ ls -l
total 44
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads
-rwxrw-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file1.txt
-rwxrw-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file2.txt
-rwx----x 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:56 Pictures
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos
kw2020202090@ubuntu:~$
```

splab 디렉토리가 삭제되었다.

- rm

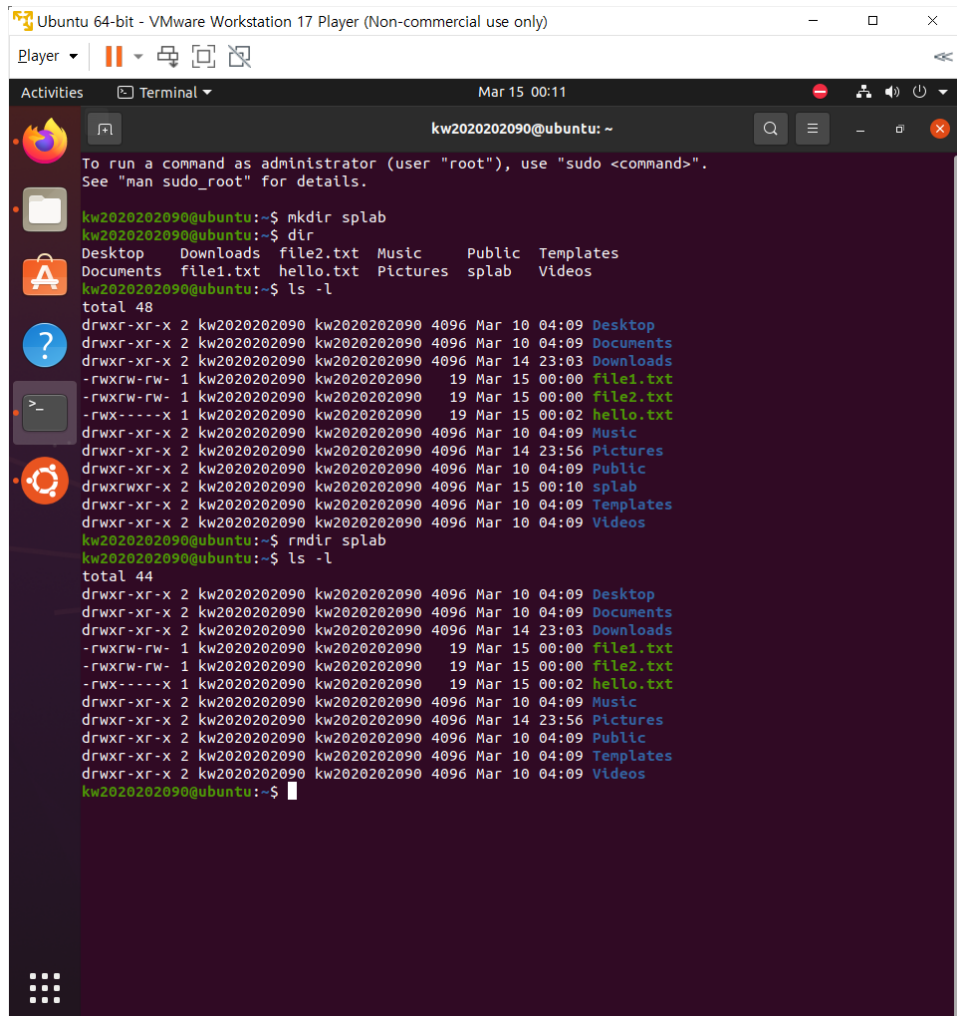
■ rm (file)



```
Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)
Player
Activities Terminal Mar 15 00:11
kw2020202090@ubuntu: ~
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
kw2020202090@ubuntu:~$ mkdir splab
kw2020202090@ubuntu:~$ dir
Desktop Downloads file2.txt Music Public Templates
Documents file1.txt hello.txt Pictures splab Videos
kw2020202090@ubuntu:~$ ls -l
total 48
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads
-rwxrwx-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file1.txt
-rwxrwx-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file2.txt
-rwx-----x 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:56 Pictures
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public
drwxrwxr-x 2 kw2020202090 kw2020202090 4096 Mar 15 00:10 splab
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos
kw2020202090@ubuntu:~$ rmdir splab
kw2020202090@ubuntu:~$ ls -l
total 44
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads
-rwxrwx-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file1.txt
-rwxrwx-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file2.txt
-rwx-----x 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:56 Pictures
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos
kw2020202090@ubuntu:~$
```

file1.txt 파일이 삭제되었다.

■ rm -r (file)

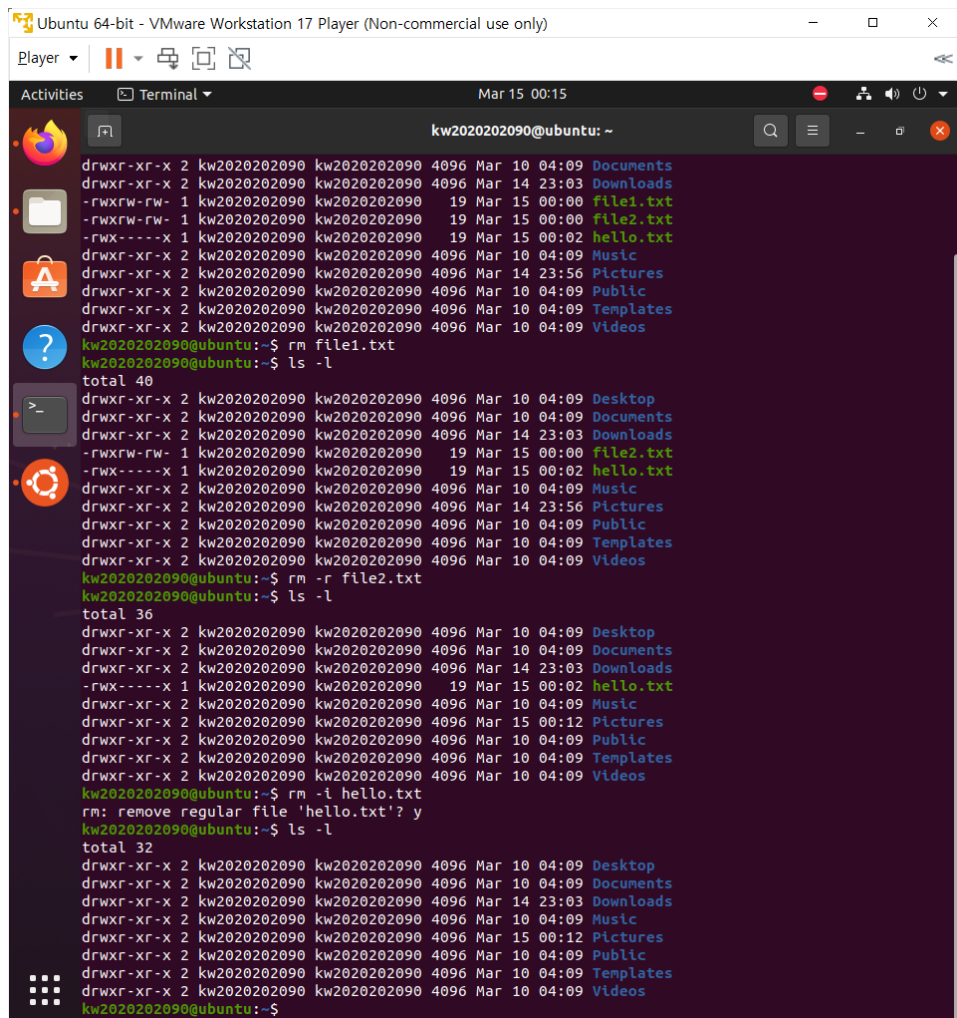


The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal output is as follows:

```
kw2020202090@ubuntu: ~  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
kw2020202090@ubuntu:~$ mkdir splab  
kw2020202090@ubuntu:~$ dir  
Desktop Downloads file2.txt Music Public Templates  
Documents file1.txt hello.txt Pictures splab Videos  
kw2020202090@ubuntu:~$ ls -l  
total 48  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads  
-rwxrwx-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file1.txt  
-rwxrwx-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file2.txt  
-rwx-----x 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:56 Pictures  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public  
drwxrwxr-x 2 kw2020202090 kw2020202090 4096 Mar 15 00:10 splab  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos  
kw2020202090@ubuntu:~$ rmdir splab  
kw2020202090@ubuntu:~$ ls -l  
total 44  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads  
-rwxrwx-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file1.txt  
-rwxrwx-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file2.txt  
-rwx-----x 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:56 Pictures  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos  
kw2020202090@ubuntu:~$
```

file2.txt 파일이 삭제되었다. (-r = recursively)

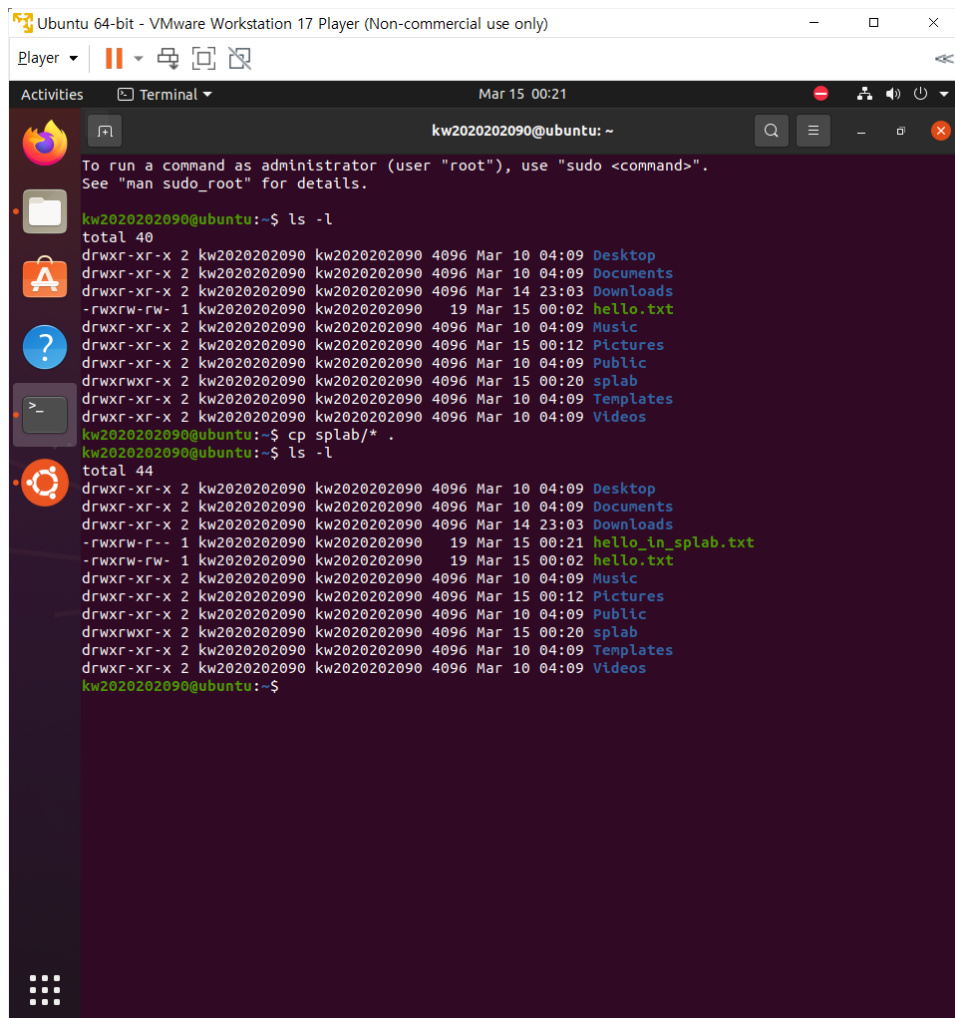
■ rm -l (file)



```
kw2020202090@ubuntu: ~  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads  
-rwxrwx-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file1.txt  
-rwxrwx-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file2.txt  
-rwx-----x 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:56 Pictures  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos  
kw2020202090@ubuntu:~$ rm file1.txt  
kw2020202090@ubuntu:~$ ls -l  
total 40  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads  
-rwxrwx-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:00 file2.txt  
-rwx-----x 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:56 Pictures  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos  
kw2020202090@ubuntu:~$ rm -r file2.txt  
kw2020202090@ubuntu:~$ ls -l  
total 36  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads  
-rwx-----x 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 15 00:12 Pictures  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos  
kw2020202090@ubuntu:~$ rm -i hello.txt  
rm: remove regular file 'hello.txt'? y  
kw2020202090@ubuntu:~$ ls -l  
total 32  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 15 00:12 Pictures  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos  
kw2020202090@ubuntu:~$
```

hello.txt를 삭제할 지에 대한 prompt를 출력하는데 y를 입력하여 hello.txt를 삭제하였다.

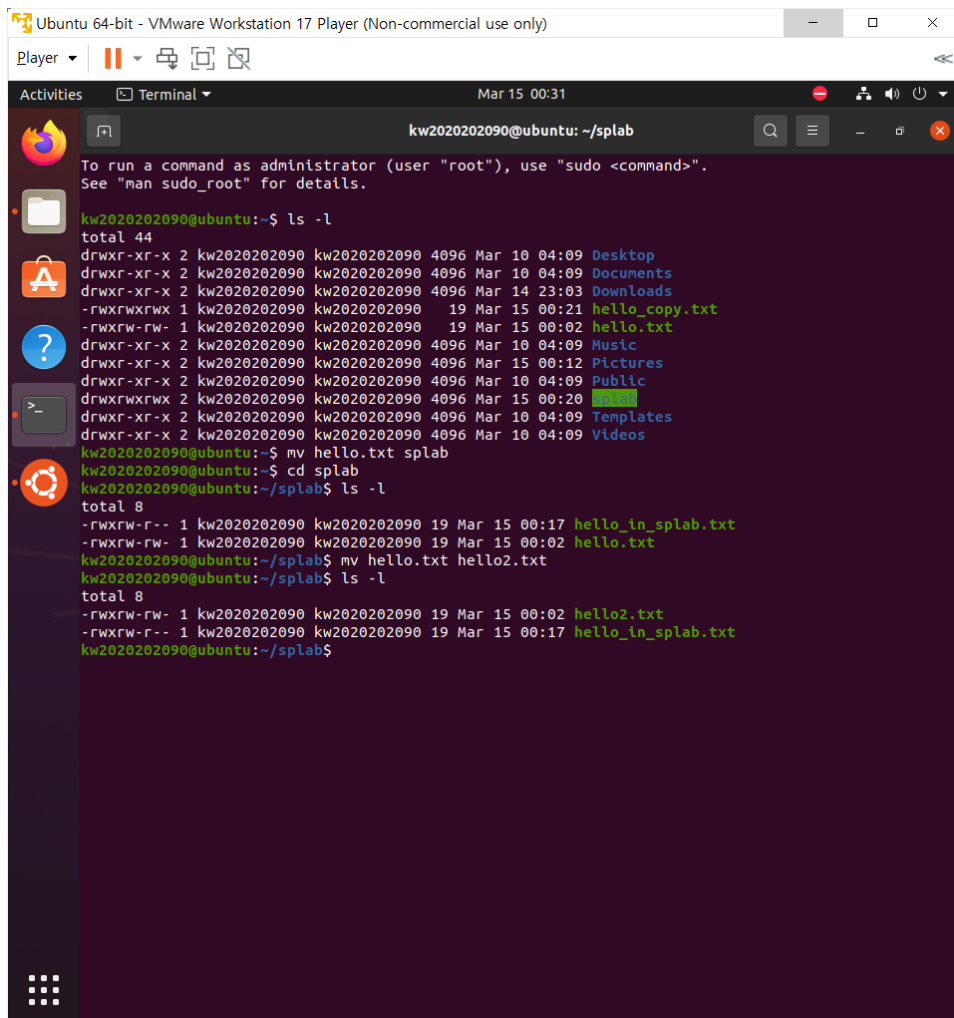
- cp



```
Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)
Player
Activities Terminal Mar 15 00:21
kw2020202090@ubuntu: ~
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
kw2020202090@ubuntu:~$ ls -l
total 40
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads
-rwxrwx-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 15 00:12 Pictures
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public
drwxrwxr-x 2 kw2020202090 kw2020202090 4096 Mar 15 00:20 splab
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos
kw2020202090@ubuntu:~$ cp splab/* .
kw2020202090@ubuntu:~$ ls -l
total 44
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads
-rwxrwx-r-- 1 kw2020202090 kw2020202090 19 Mar 15 00:21 hello_in_splab.txt
-rwxrwx-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 15 00:12 Pictures
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public
drwxrwxr-x 2 kw2020202090 kw2020202090 4096 Mar 15 00:20 splab
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos
kw2020202090@ubuntu:~$
```

hello.txt 파일을 hello_copy.txt라는 이름으로 복사하여 생성하였다.
혹은 cp (directory1)/* (directory2)를 사용하면 directory1의 모든 파일을
directory 2로 복사하여 생성할 수 있다.

- mv

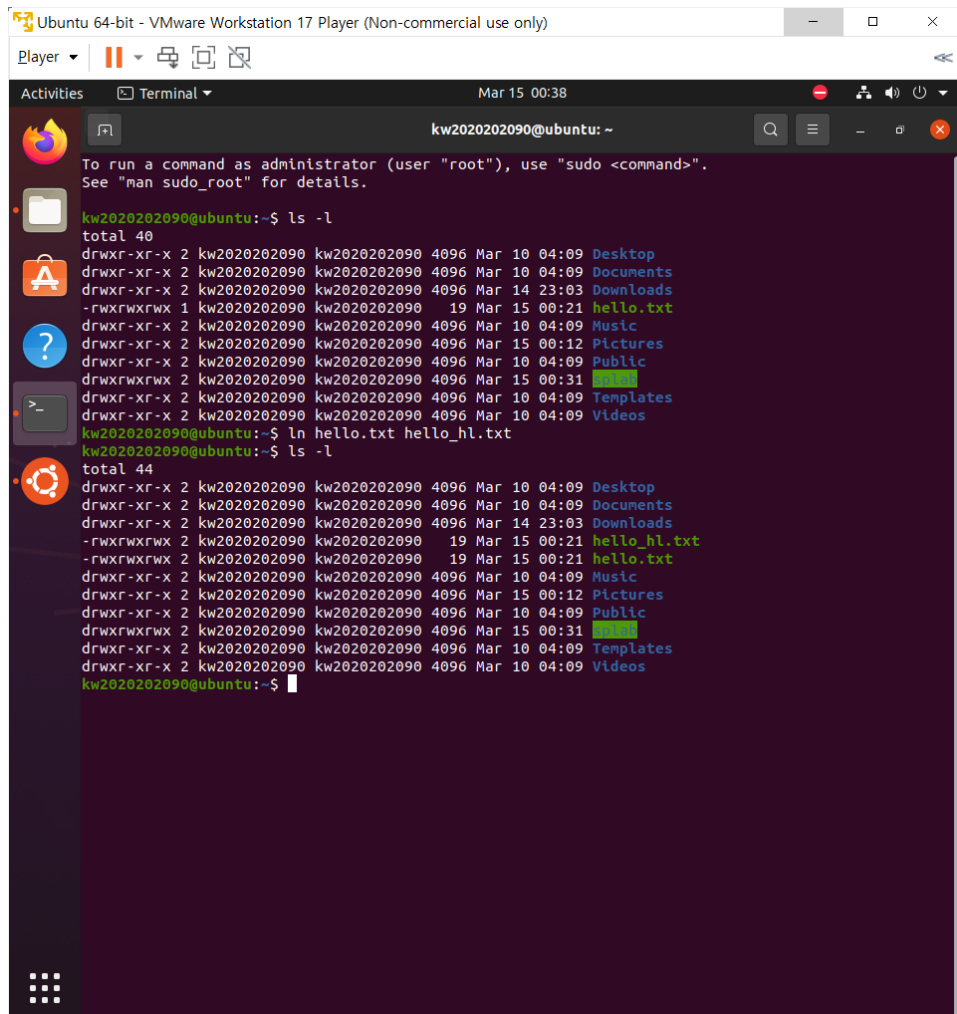


```
Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)
Player
Activities Terminal Mar 15 00:31
kw2020202090@ubuntu: ~/splab
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
kw2020202090@ubuntu:~$ ls -l
total 44
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads
-rwxrwxrwx 1 kw2020202090 kw2020202090 19 Mar 15 00:21 hello_copy.txt
-rwxrw-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 15 00:12 Pictures
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public
drwxrwxrwx 2 kw2020202090 kw2020202090 4096 Mar 15 00:20 splab
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos
kw2020202090@ubuntu:~$ mv hello.txt splab
kw2020202090@ubuntu:~$ cd splab
kw2020202090@ubuntu:~/splab$ ls -l
total 8
-rwxrw-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:17 hello_in_splab.txt
-rwxrw-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello.txt
kw2020202090@ubuntu:~/splab$ mv hello.txt hello2.txt
kw2020202090@ubuntu:~/splab$ ls -l
total 8
-rwxrw-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:02 hello2.txt
-rwxrw-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:17 hello_in_splab.txt
kw2020202090@ubuntu:~/splab$
```

파일을 다른 디렉토리로 이동하거나 파일의 이름을 변경할 수 있다.

- ln

■ ln (file1) (hard_link_name)

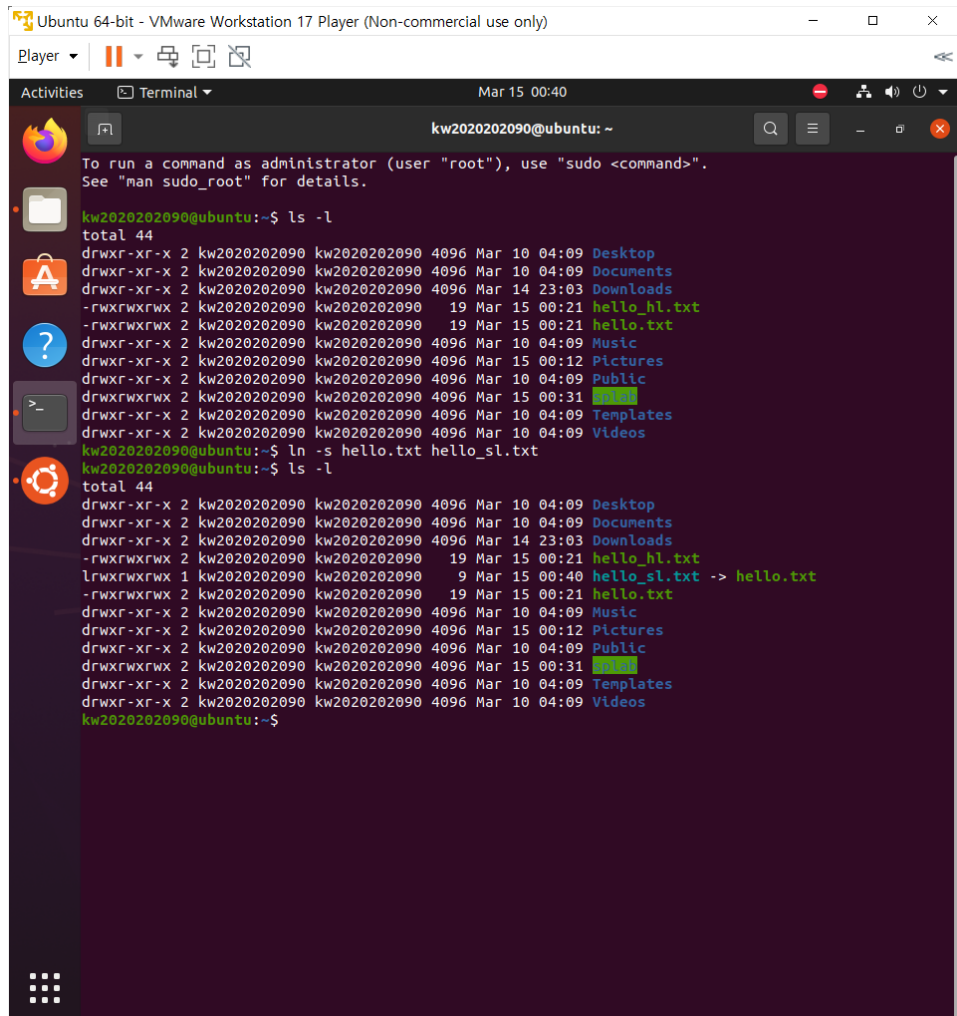


The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal output is as follows:

```
kw2020202090@ubuntu: ~  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
kw2020202090@ubuntu:~$ ls -l  
total 40  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads  
-rw-rw-rw- 1 kw2020202090 kw2020202090 19 Mar 15 00:21 hello.txt  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 15 00:12 Pictures  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public  
drwxrwxrwx 2 kw2020202090 kw2020202090 4096 Mar 15 00:31 snap  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos  
kw2020202090@ubuntu:~$ ln hello.txt hello_hl.txt  
kw2020202090@ubuntu:~$ ls -l  
total 44  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads  
-rw-rw-rw- 2 kw2020202090 kw2020202090 19 Mar 15 00:21 hello_hl.txt  
-rw-rw-rw- 2 kw2020202090 kw2020202090 19 Mar 15 00:21 hello.txt  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 15 00:12 Pictures  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public  
drwxrwxrwx 2 kw2020202090 kw2020202090 4096 Mar 15 00:31 snap  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos  
kw2020202090@ubuntu:~$
```

hello.txt 파일에 대한 hello_hl.txt라는 하드링크 파일을 생성했다.

■ ln -s (file) (symbolic_link_name)

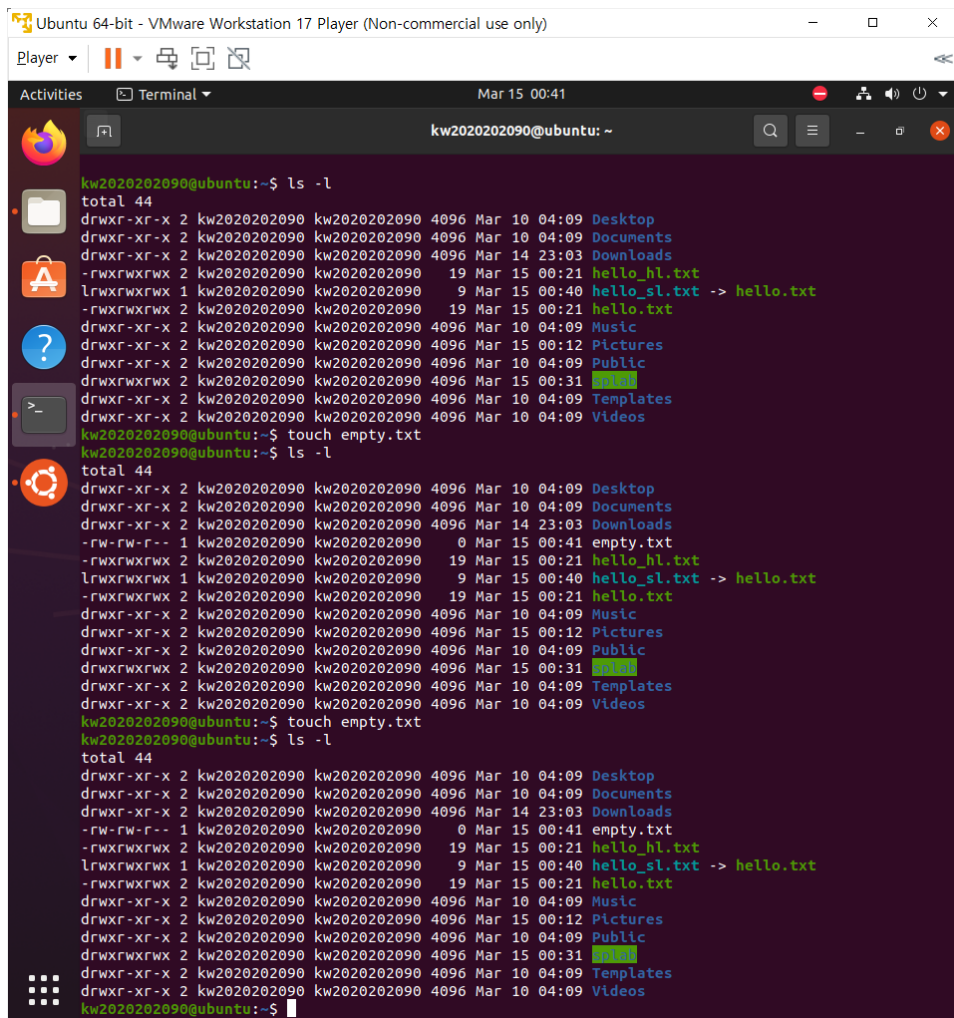


The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal output is as follows:

```
kw2020202090@ubuntu: ~  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
kw2020202090@ubuntu:~$ ls -l  
total 44  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads  
-rw-rw-rw- 2 kw2020202090 kw2020202090 19 Mar 15 00:21 hello_hl.txt  
-rw-rw-rw- 2 kw2020202090 kw2020202090 19 Mar 15 00:21 hello.txt  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 15 00:12 Pictures  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public  
drwxrwxrwx 2 kw2020202090 kw2020202090 4096 Mar 15 00:31 tmp  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos  
kw2020202090@ubuntu:~$ ln -s hello.txt hello_sl.txt  
kw2020202090@ubuntu:~$ ls -l  
total 44  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads  
-rw-rw-rw- 2 kw2020202090 kw2020202090 19 Mar 15 00:21 hello_hl.txt  
lrwxrwxrwx 1 kw2020202090 kw2020202090 9 Mar 15 00:40 hello_sl.txt -> hello.txt  
-rw-rw-rw- 2 kw2020202090 kw2020202090 19 Mar 15 00:21 hello.txt  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 15 00:12 Pictures  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public  
drwxrwxrwx 2 kw2020202090 kw2020202090 4096 Mar 15 00:31 tmp  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates  
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos  
kw2020202090@ubuntu:~$
```

hello.txt 파일에 대한 hello_sl.txt라는 심볼릭링크 파일을 생성했다.

- touch



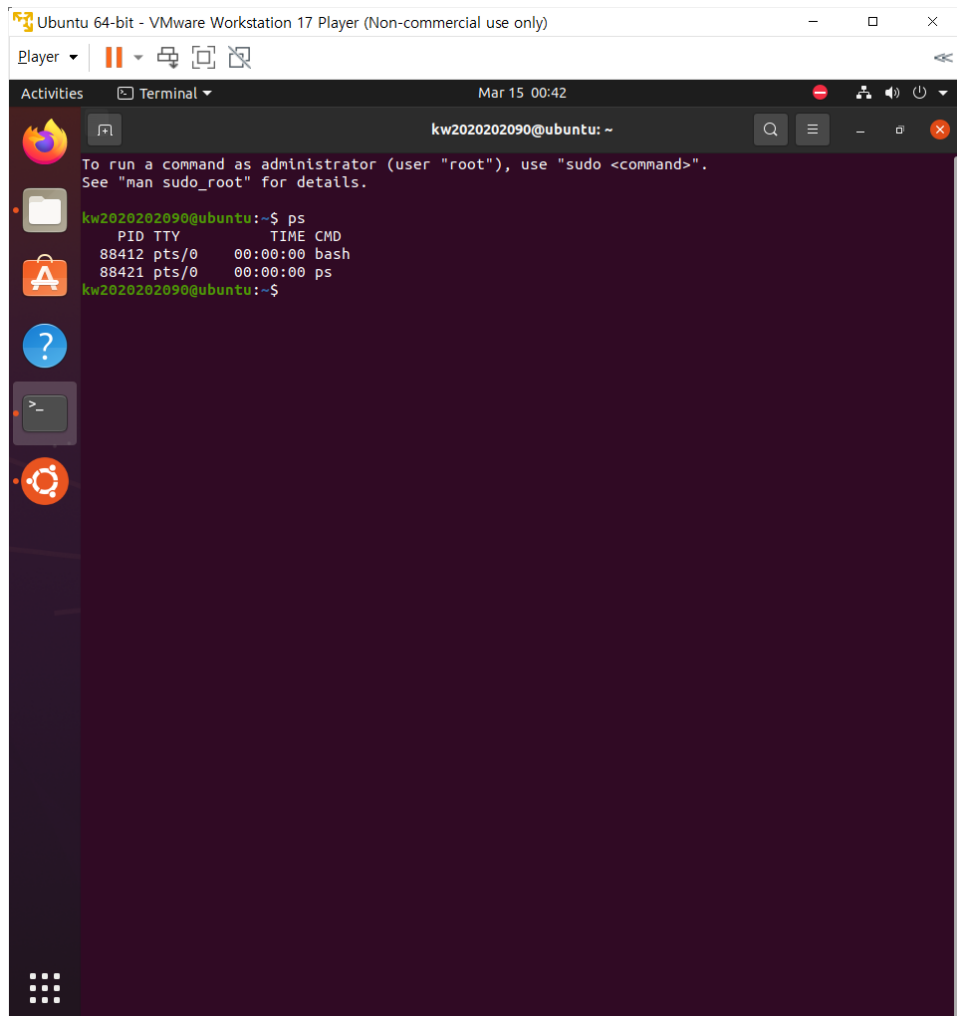
The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal output is as follows:

```
kw2020202090@ubuntu:~$ ls -l
total 44
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads
-rwxrwxrwx 1 kw2020202090 kw2020202090 19 Mar 15 00:21 hello_hl.txt
lrwxrwxrwx 1 kw2020202090 kw2020202090 9 Mar 15 00:40 hello_sl.txt -> hello.txt
-rwxrwxrwx 1 kw2020202090 kw2020202090 19 Mar 15 00:21 hello.txt
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 15 00:12 Pictures
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public
drwxrwxrwx 1 kw2020202090 kw2020202090 4096 Mar 15 00:31 srtan
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos
kw2020202090@ubuntu:~$ touch empty.txt
kw2020202090@ubuntu:~$ ls -l
total 44
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads
-rw-rw-r-- 1 kw2020202090 kw2020202090 0 Mar 15 00:41 empty.txt
-rwxrwxrwx 1 kw2020202090 kw2020202090 19 Mar 15 00:21 hello_hl.txt
lrwxrwxrwx 1 kw2020202090 kw2020202090 9 Mar 15 00:40 hello_sl.txt -> hello.txt
-rwxrwxrwx 1 kw2020202090 kw2020202090 19 Mar 15 00:21 hello.txt
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 15 00:12 Pictures
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public
drwxrwxrwx 1 kw2020202090 kw2020202090 4096 Mar 15 00:31 srtan
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos
kw2020202090@ubuntu:~$ touch empty.txt
kw2020202090@ubuntu:~$ ls -l
total 44
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Desktop
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Documents
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 14 23:03 Downloads
-rw-rw-r-- 1 kw2020202090 kw2020202090 0 Mar 15 00:41 empty.txt
-rwxrwxrwx 1 kw2020202090 kw2020202090 19 Mar 15 00:21 hello_hl.txt
lrwxrwxrwx 1 kw2020202090 kw2020202090 9 Mar 15 00:40 hello_sl.txt -> hello.txt
-rwxrwxrwx 1 kw2020202090 kw2020202090 19 Mar 15 00:21 hello.txt
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Music
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 15 00:12 Pictures
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Public
drwxrwxrwx 1 kw2020202090 kw2020202090 4096 Mar 15 00:31 srtan
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Templates
drwxr-xr-x 2 kw2020202090 kw2020202090 4096 Mar 10 04:09 Videos
kw2020202090@ubuntu:~$
```

empty.txt라는 빈 파일을 생성한다.

- ps

■ ps

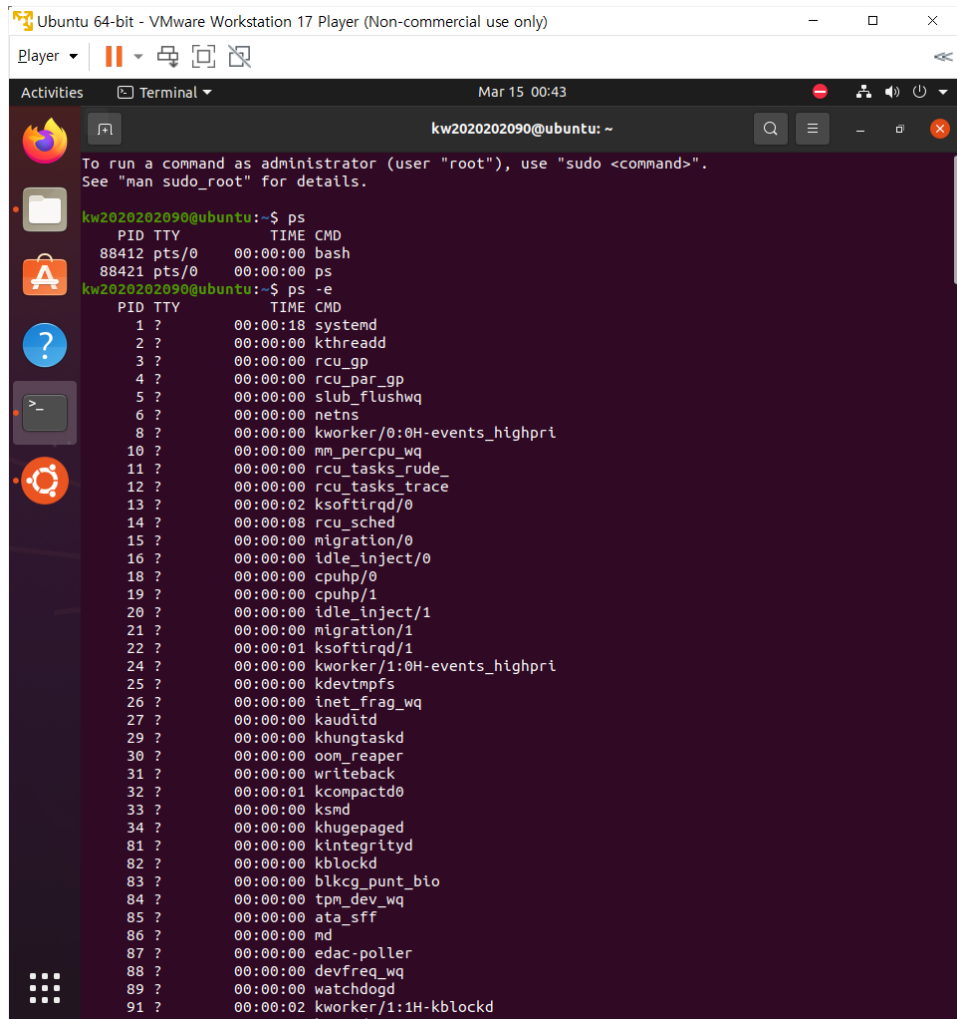


The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal prompt is "kw2020202090@ubuntu: ~". The user has entered the command "ps", and the output is as follows:

```
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
kw2020202090@ubuntu:~$ ps  
  PID TTY          TIME CMD  
 88412 pts/0    00:00:00 bash  
 88421 pts/0    00:00:00 ps  
kw2020202090@ubuntu:~$
```

현재 프로세스 상태를 출력한다.

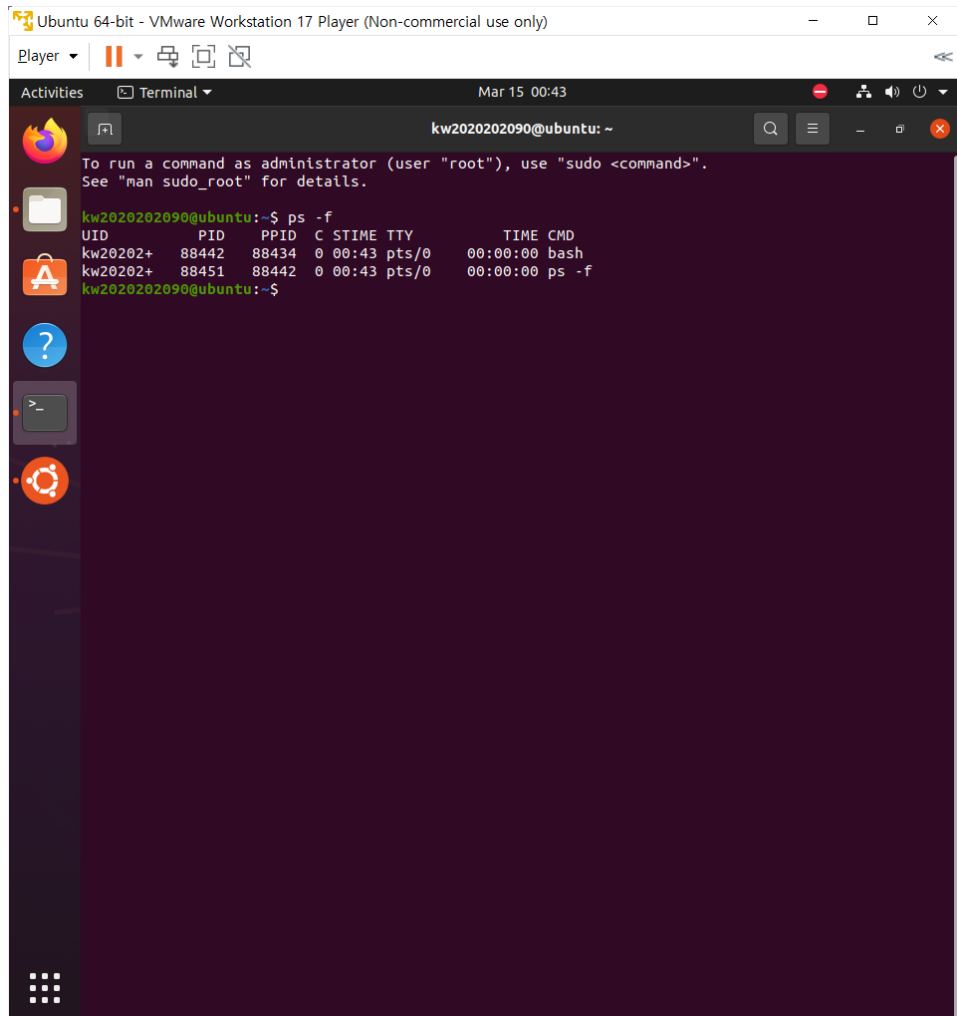
■ ps -e



```
kw2020202090@ubuntu: ~  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
kw2020202090@ubuntu:~$ ps  
  PID TTY          TIME CMD  
 88412 pts/0    00:00:00 bash  
 88421 pts/0    00:00:00 ps  
kw2020202090@ubuntu:~$ ps -e  
  PID TTY          TIME CMD  
    1 ?        00:00:18 systemd  
    2 ?        00:00:00 kthreadd  
    3 ?        00:00:00 rcu_gp  
    4 ?        00:00:00 rcu_par_gp  
    5 ?        00:00:00 slub_flushwq  
    6 ?        00:00:00 netns  
    8 ?        00:00:00 kworker/0:0H-events_highpri  
   10 ?        00:00:00 mm_percpu_wq  
   11 ?        00:00:00 rcu_tasks_rude_  
   12 ?        00:00:00 rcu_tasks_trace  
   13 ?        00:00:02 ksoftirqd/0  
   14 ?        00:00:08 rcu_sched  
   15 ?        00:00:00 migration/0  
   16 ?        00:00:00 idle_inject/0  
   18 ?        00:00:00 cpuphp/0  
   19 ?        00:00:00 cpuphp/1  
   20 ?        00:00:00 idle_inject/1  
   21 ?        00:00:00 migration/1  
   22 ?        00:00:01 ksoftirqd/1  
   24 ?        00:00:00 kworker/1:0H-events_highpri  
   25 ?        00:00:00 kdevtmpfs  
   26 ?        00:00:00 inet_frag_wq  
   27 ?        00:00:00 kauditd  
   29 ?        00:00:00 khungtaskd  
   30 ?        00:00:00 oom_reaper  
   31 ?        00:00:00 writeback  
   32 ?        00:00:01 kcompactd0  
   33 ?        00:00:00 ksm  
   34 ?        00:00:00 khugepaged  
   81 ?        00:00:00 kintegrityd  
   82 ?        00:00:00 kblockd  
   83 ?        00:00:00 blkcg_punt_bio  
   84 ?        00:00:00 tpm_dev_wq  
   85 ?        00:00:00 ata_sff  
   86 ?        00:00:00 md  
   87 ?        00:00:00 edac-poller  
   88 ?        00:00:00 devfreq_wq  
   89 ?        00:00:00 watchdogd  
   91 ?        00:00:02 kworker/1:1H-kblockd
```

현재 모든 프로세스 상태를 출력한다.

■ ps -f



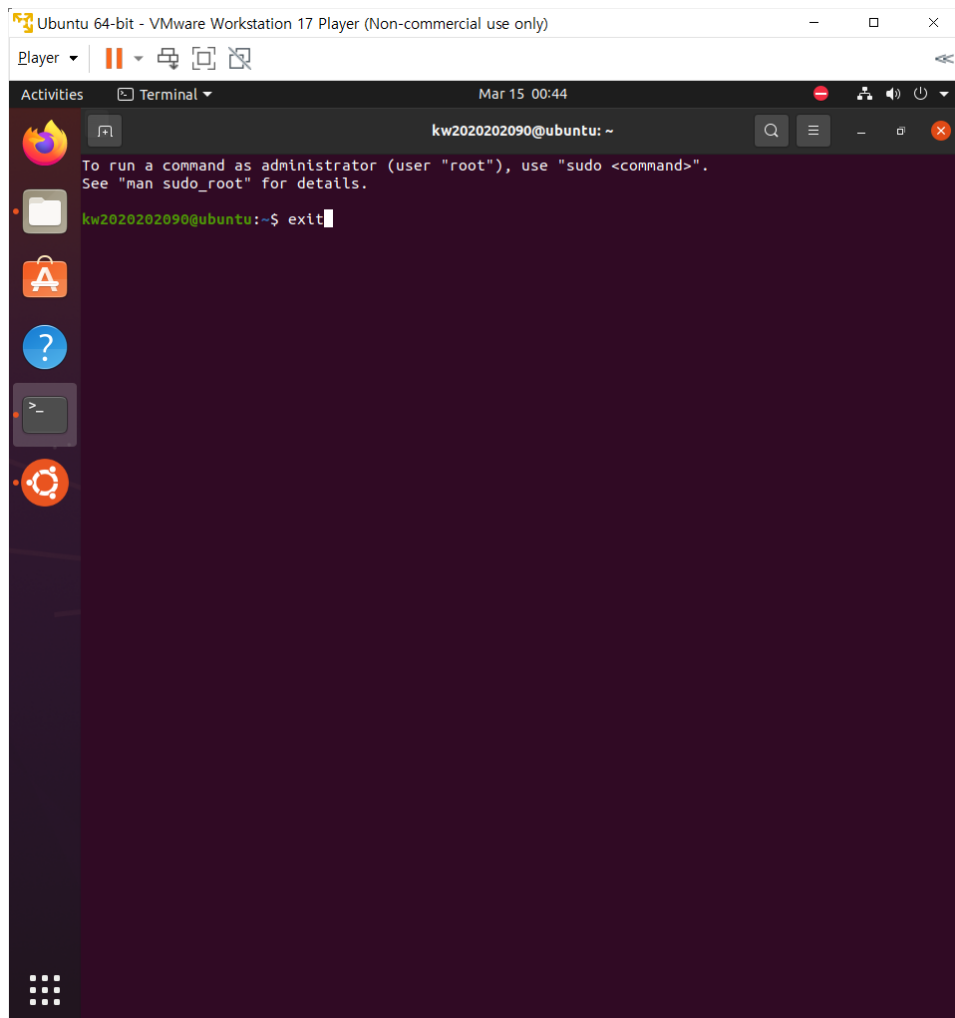
The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal is running the command `ps -f`. The output displays the following table:

UID	PID	PPID	C	STIME	TTY	TIME	CMD
kw20202+	88442	88434	0	00:43	pts/0	00:00:00	bash
kw20202+	88451	88442	0	00:43	pts/0	00:00:00	ps -f

The terminal prompt is `kw20202090@ubuntu: ~`. The left sidebar of the terminal window shows various application icons, including the Dash icon, Home icon, Files icon, Applications icon, and the Dash icon.

현재 프로세스 상태를 자세히 출력한다.

- exit



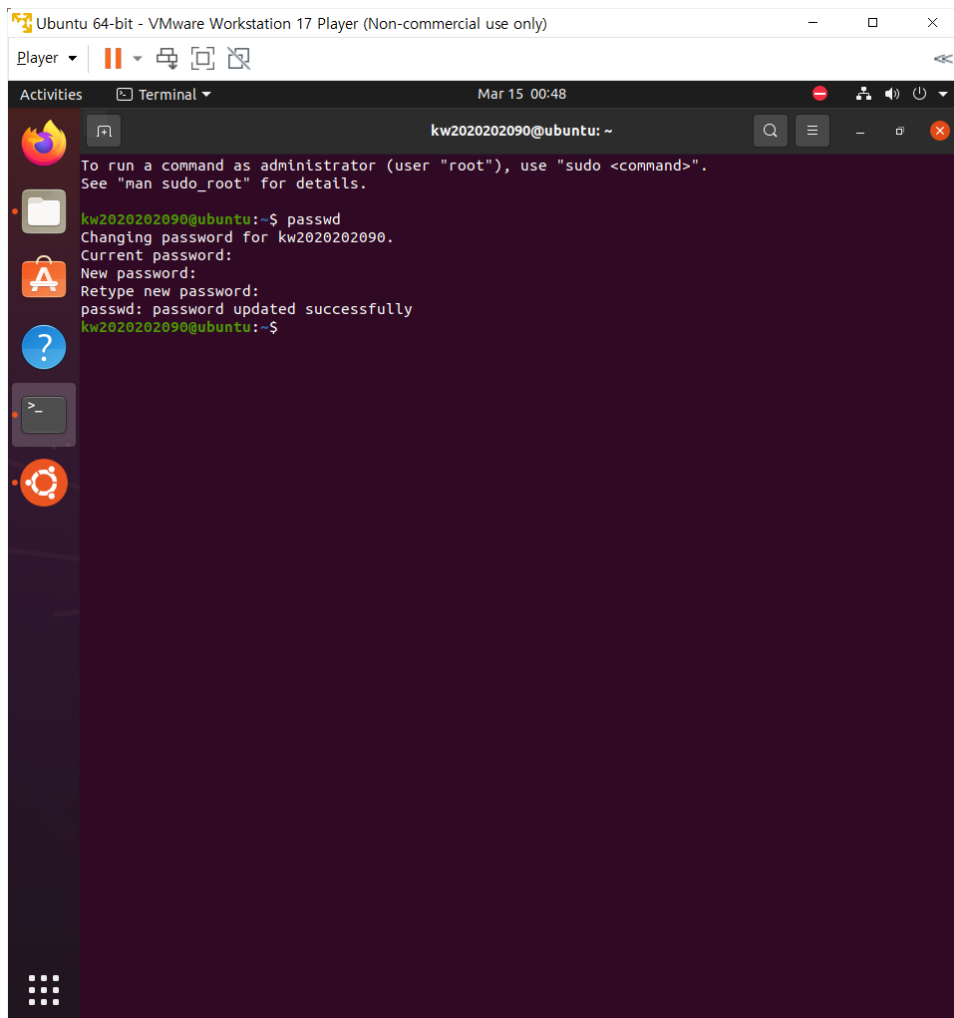
터미널에 exit를 입력하면 터미널을 종료한다.

- kill

```
kw2020202090@ubuntu: ~  
1799 ?      00:00:00 gsd-disk-utilit  
1800 ?      00:00:00 gsd-power  
1801 ?      00:00:00 gsd-print-notif  
1805 ?      00:00:00 gsd-rfkill  
1808 ?      00:00:00 gsd-screensaver  
1818 ?      00:00:00 gsd-sharing  
1825 ?      00:00:00 gsd-smartcard  
1833 ?      00:00:00 gsd-sound  
1836 ?      00:00:00 gsd-usb-protect  
1839 ?      00:00:00 evolution-alarm  
1846 ?      00:00:00 vmttoolsd  
1847 ?      00:00:00 gsd-wacom  
1850 ?      00:00:00 gsd-wwan  
1851 ?      00:00:00 gsd-xsettings  
1856 ?      00:00:00 ibus-engine-sim  
1901 ?      00:00:00 gsd-printer  
1934 ?      00:00:01 gnome-initial-s  
1975 ?      00:00:31 gnome-terminal-  
1983 pts/0   00:00:00 bash  
1993 pts/0   00:00:11 yes  
2011 pts/1   00:00:00 bash  
2029 pts/1   00:00:00 ps  
kw2020202090@ubuntu:~$ kill 1993  
kw2020202090@ubuntu:~$  
Terminated  
kw2020202090@ubuntu:~$
```

my name을 무한 출력하는 yes 프로세스를 종료하였다.

- passwd



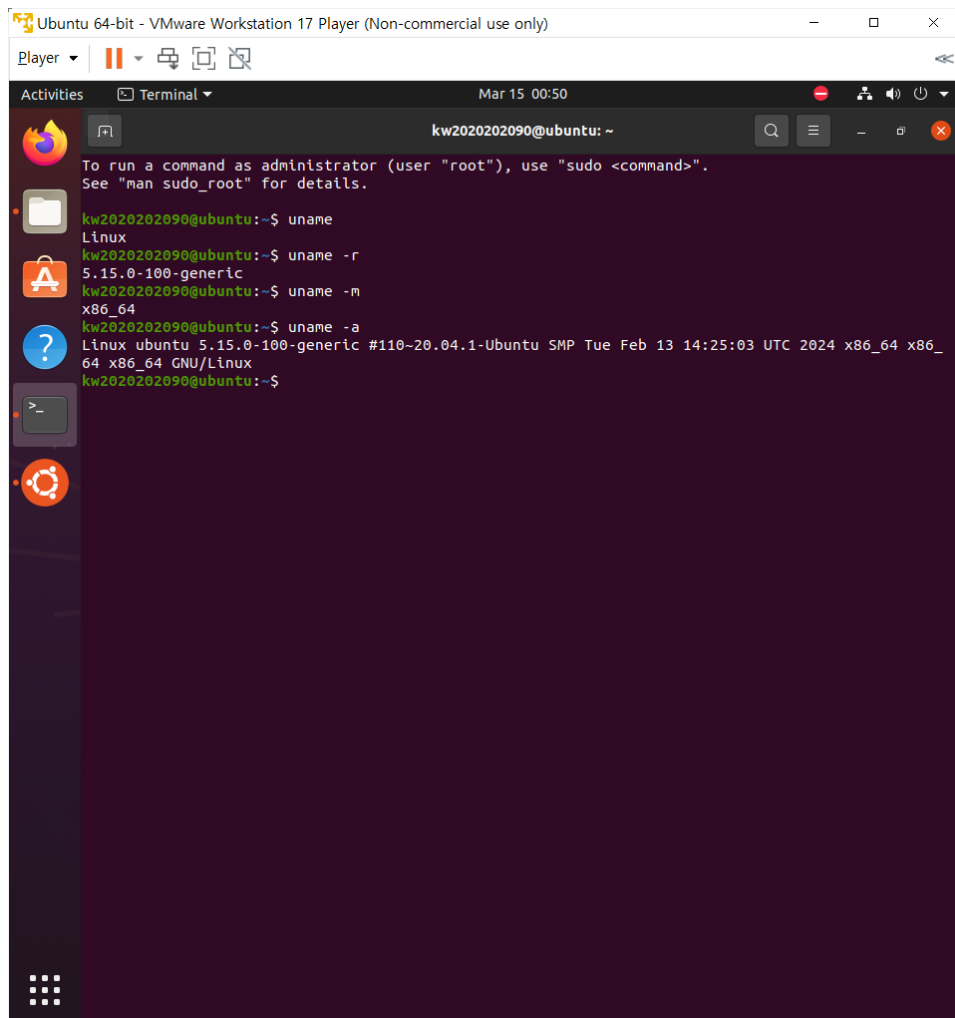
The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal prompt is "kw2020202090@ubuntu: ~". The user has entered the command "passwd". The terminal output is as follows:

```
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

kw2020202090@ubuntu:~$ passwd
Changing password for kw2020202090.
Current password:
New password:
Retype new password:
passwd: password updated successfully
kw2020202090@ubuntu:~$
```

사용자의 비밀번호를 변경할 수 있다.

- uname

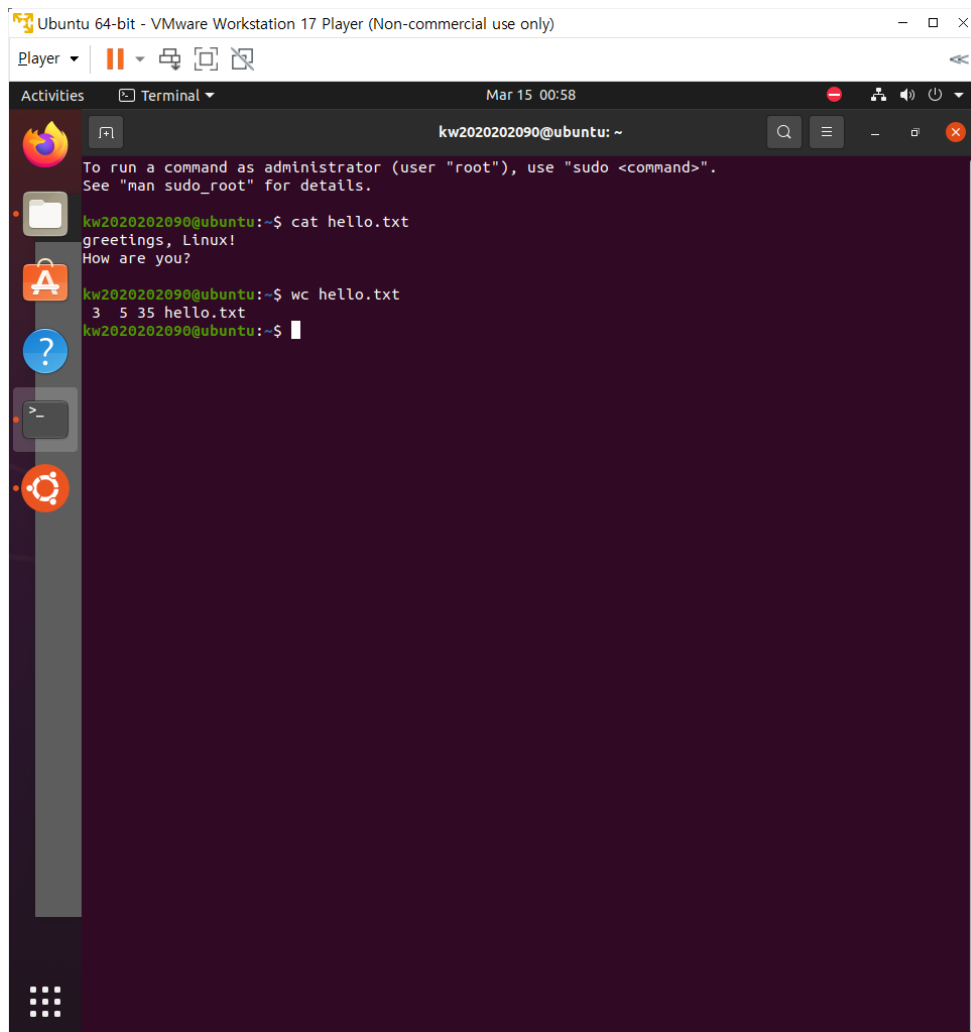


The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal is running on a system with the username "kw2020202090@ubuntu". The prompt is "kw2020202090@ubuntu: ~". The terminal displays the output of the 'uname' command and its various options:

```
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
kw2020202090@ubuntu:~$ uname  
Linux  
kw2020202090@ubuntu:~$ uname -r  
5.15.0-100-generic  
kw2020202090@ubuntu:~$ uname -m  
x86_64  
kw2020202090@ubuntu:~$ uname -a  
Linux ubuntu 5.15.0-100-generic #110~20.04.1-Ubuntu SMP Tue Feb 13 14:25:03 UTC 2024 x86_64 x86_64 GNU/Linux  
kw2020202090@ubuntu:~$
```

사용자의 정보를 출력한다.

- WC

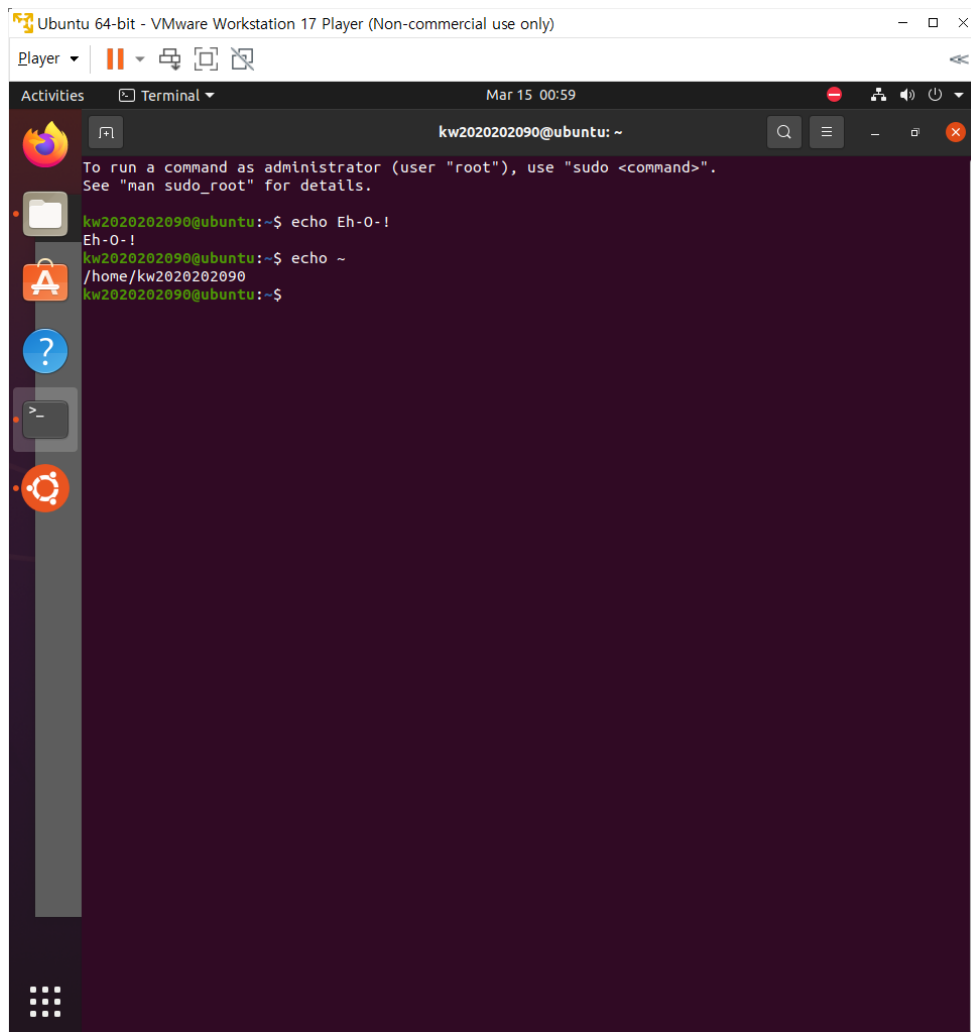


The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal prompt is "kw2020202090@ubuntu: ~". The user has entered the command "cat hello.txt", which outputs "greetings, Linux!" and "How are you?". The user then enters the command "wc hello.txt", which outputs "3 5 35 hello.txt". The terminal window has a dark purple background and a light gray border. The VMware Workstation window has a title bar with standard window controls and a menu bar with "Player", "Terminal", and "Mar 15 00:58".

```
kw2020202090@ubuntu: ~  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
kw2020202090@ubuntu:~$ cat hello.txt  
greetings, Linux!  
How are you?  
kw2020202090@ubuntu:~$ wc hello.txt  
3 5 35 hello.txt  
kw2020202090@ubuntu:~$
```

hello.txt 파일의 개행 수, 단어 수, 바이트 수를 출력한다.

- echo

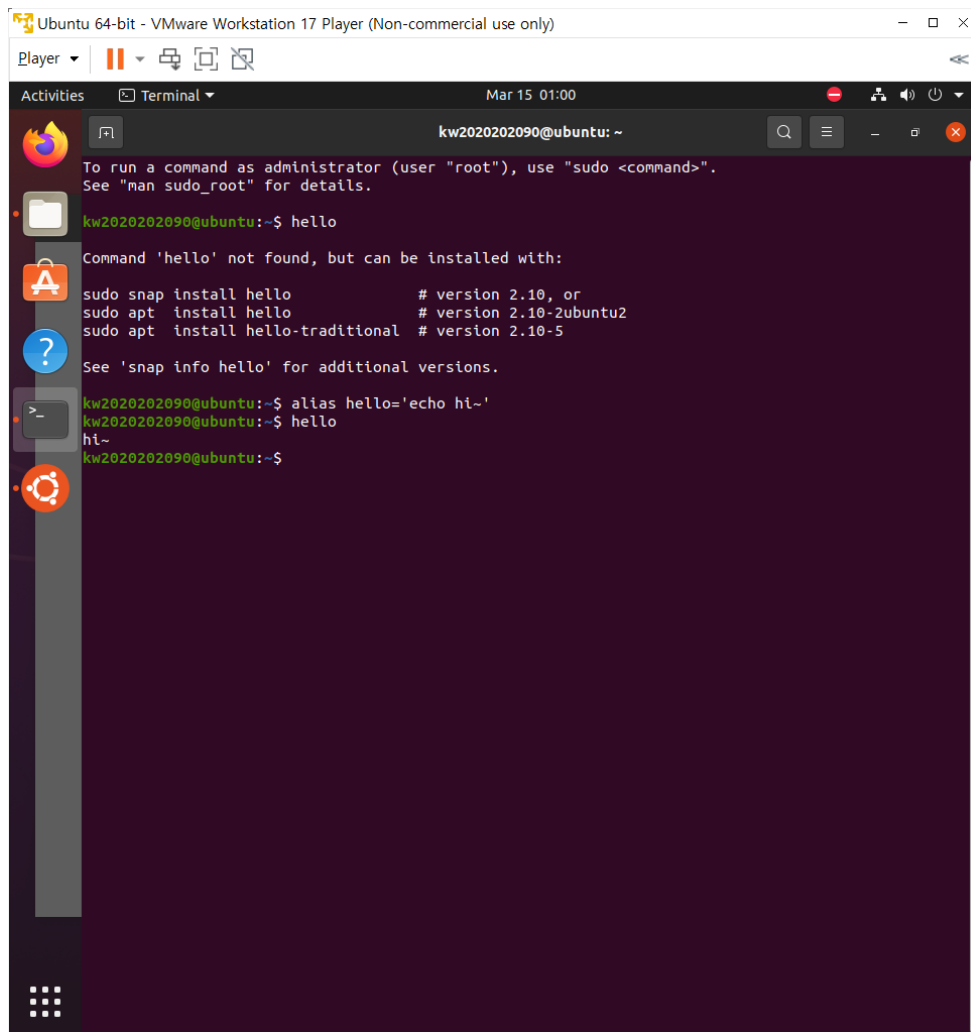


The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal is running on a system with the username "kw2020202090@ubuntu". The prompt is "kw2020202090@ubuntu: ~". The terminal displays the following commands and output:

```
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
kw2020202090@ubuntu:~$ echo Eh-O-!  
Eh-O-!  
kw2020202090@ubuntu:~$ echo ~  
/home/kw2020202090  
kw2020202090@ubuntu:~$
```

입력한 문자열을 출력한다. 혹은 현재 환경 변수를 출력한다.

- alias



```
Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)
Player
Activities Terminal Mar 15 01:00 kw2020202090@ubuntu: ~

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

kw2020202090@ubuntu:~$ hello

Command 'hello' not found, but can be installed with:

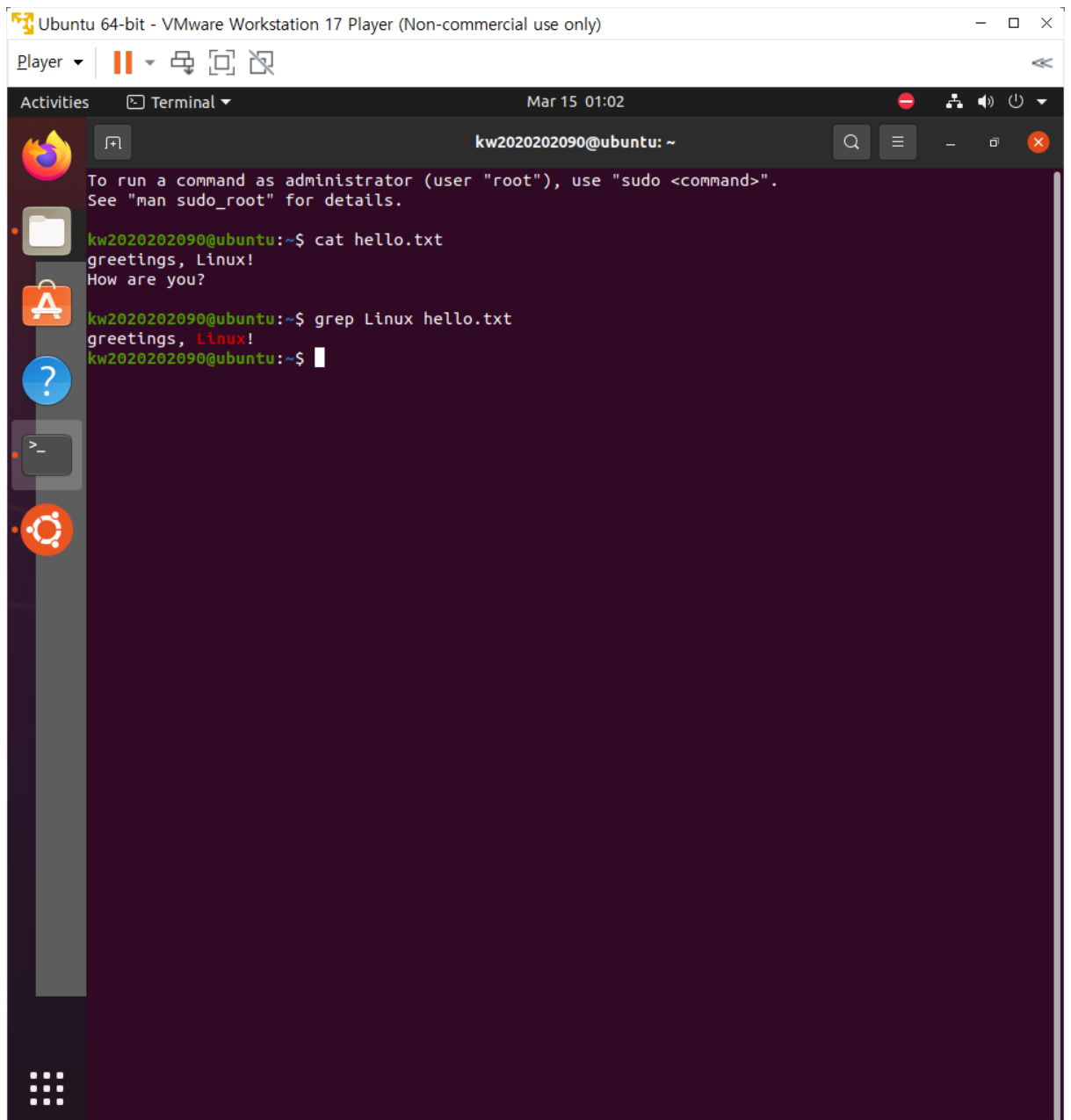
sudo snap install hello          # version 2.10, or
sudo apt install hello           # version 2.10-2ubuntu2
sudo apt install hello-traditional # version 2.10-5

See 'snap info hello' for additional versions.

kw2020202090@ubuntu:~$ alias hello='echo hi~'
kw2020202090@ubuntu:~$ hello
hi~
kw2020202090@ubuntu:~$
```

특정한 명령어에 명령어를 바인드할 수 있다.

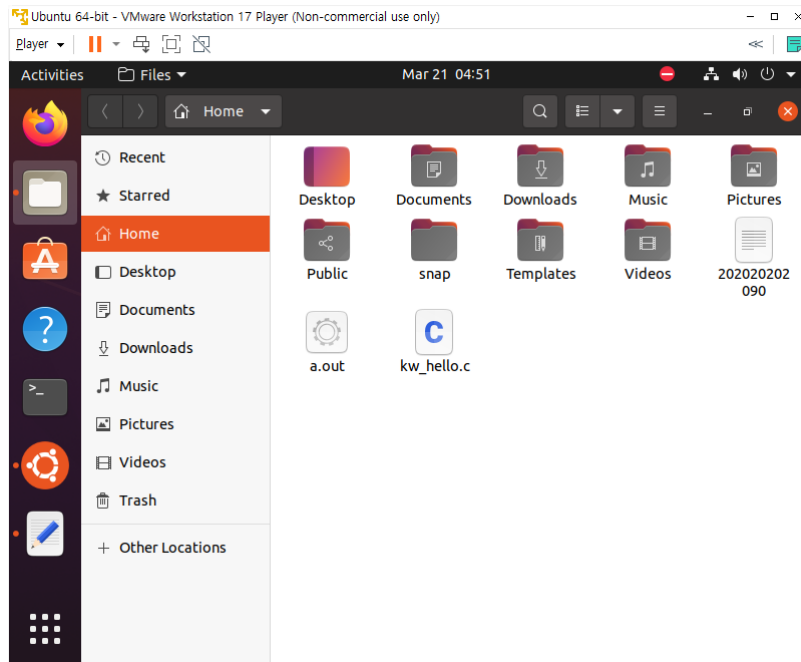
- grep



The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal output is as follows:

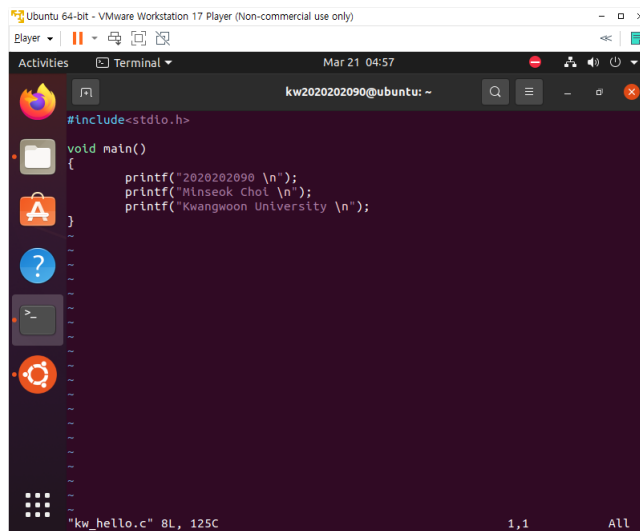
```
kw2020202090@ubuntu: ~  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
kw2020202090@ubuntu:~$ cat hello.txt  
greetings, Linux!  
How are you?  
kw2020202090@ubuntu:~$ grep Linux hello.txt  
greetings, Linux!  
kw2020202090@ubuntu:~$
```

해당 파일에서 입력된 패턴을 찾아 강조하여 출력한다.



- Make

1. vi editor로 kw_hello.c 파일 작성

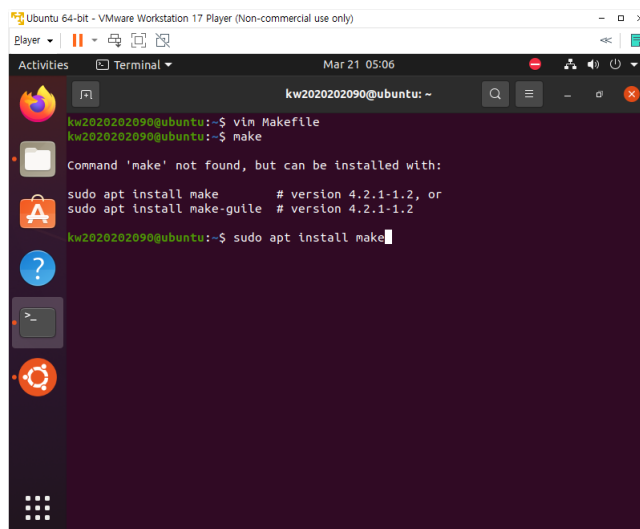


The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The user is logged in as kw2020202090@ubuntu. The terminal shows the user opening the vi editor to create a file named kw_hello.c. The file content is as follows:

```
#include<stdio.h>

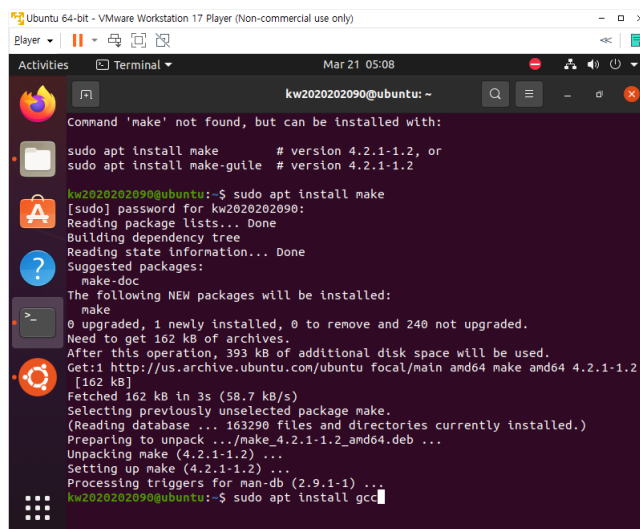
void main()
{
    printf("2020202090 \n");
    printf("Minseok Choi \n");
    printf("Kwangwoon University \n");
}
```

2. gcc, make 설치



The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The user is logged in as kw2020202090@ubuntu. The terminal shows the user attempting to run the make command, which results in an error: "Command 'make' not found, but can be installed with:". The user then runs the command "sudo apt install make", which prompts for a password and shows the installation progress.

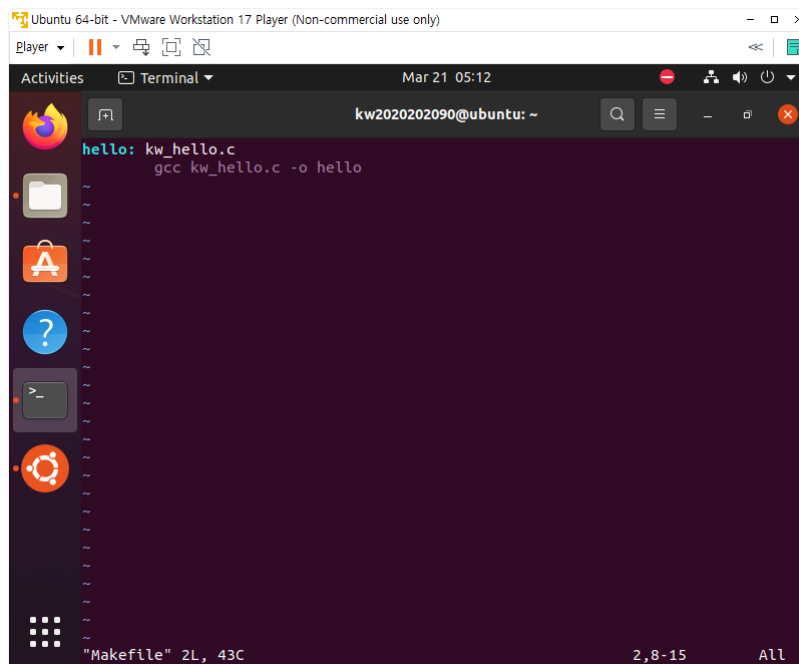
```
kw2020202090@ubuntu:~$ vim Makefile
kw2020202090@ubuntu:~$ make
Command 'make' not found, but can be installed with:
sudo apt install make      # version 4.2.1-1.2, or
sudo apt install make-guile # version 4.2.1-1.2
kw2020202090@ubuntu:~$ sudo apt install make
```



The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The user is logged in as kw2020202090@ubuntu. The terminal shows the user running the command "sudo apt install gcc", which prompts for a password and shows the installation progress.

```
kw2020202090@ubuntu:~$ sudo apt install make
[sudo] password for kw2020202090:
Reading package lists... Done
Building dependency tree
Reading state information... Done
Suggested packages:
  make-doc
The following NEW packages will be installed:
  make
0 upgraded, 1 newly installed, 0 to remove and 240 not upgraded.
Need to get 162 kB of archives.
After this operation, 393 kB of additional disk space will be used.
Get:: http://us.archive.ubuntu.com/ubuntu focal/main amd64 make amd64 4.2.1-1.2
[162 kB]
Fetched 162 kB in 3s (58.7 kB/s)
Selecting previously unselected package make.
(Reading database ... 163290 files and directories currently installed.)
Preparing to unpack .../make_4.2.1-1.2_amd64.deb ...
Unpacking make (4.2.1-1.2) ...
Setting up make (4.2.1-1.2) ...
Processing triggers for man-db (2.9.1-1) ...
kw2020202090@ubuntu:~$ sudo apt install gcc
```

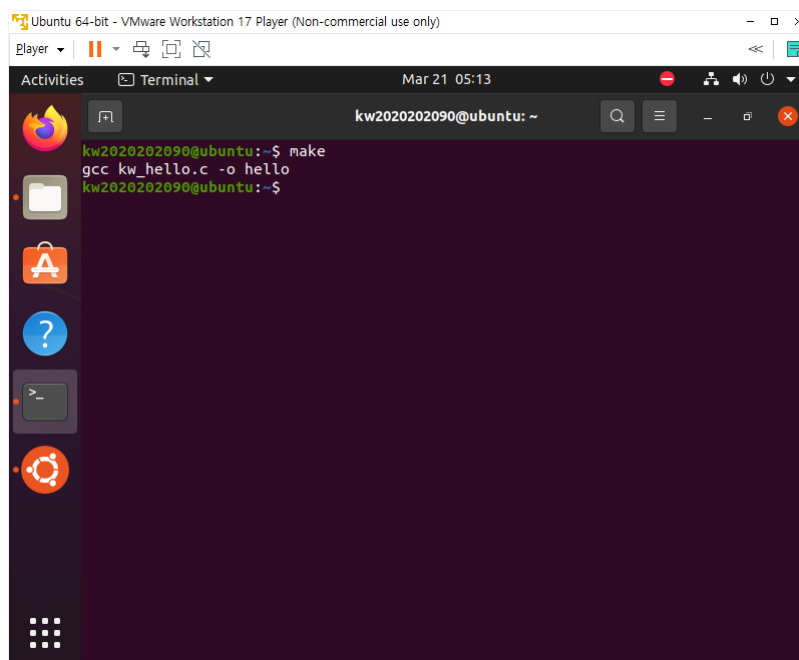
3. makefile 작성



The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal prompt is "kw2020202090@ubuntu: ~". The user has entered the command "hello: kw_hello.c" followed by "gcc kw_hello.c -o hello". The terminal output shows the compilation process. The status bar at the bottom indicates "Makefile" 2L, 43C, 2,8-15, and All.

```
kw2020202090@ubuntu: ~  
hello: kw_hello.c  
gcc kw_hello.c -o hello
```

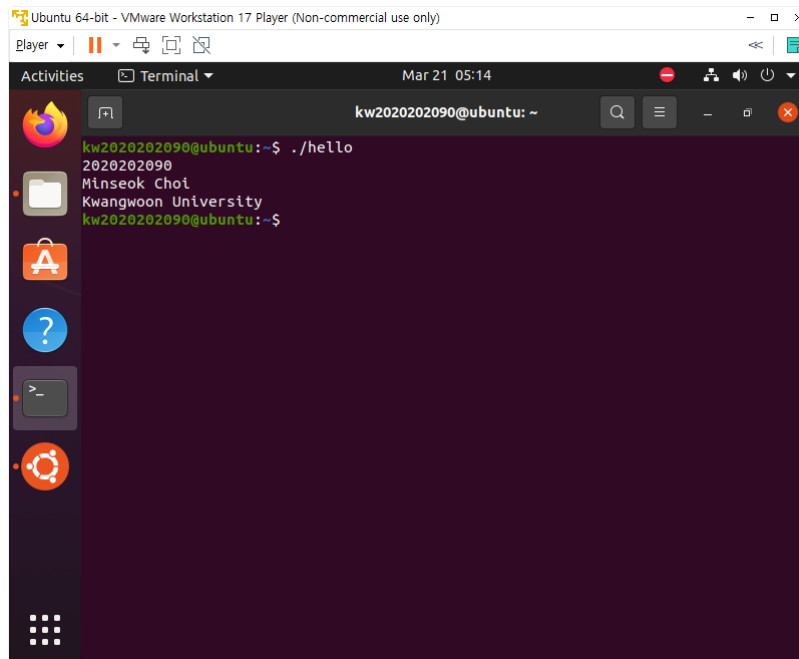
4. make



The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal prompt is "kw2020202090@ubuntu: ~". The user has entered the command "make". The terminal output shows the compilation process. The status bar at the bottom indicates "Makefile" 2L, 43C, 2,8-15, and All.

```
kw2020202090@ubuntu: ~$ make  
gcc kw_hello.c -o hello  
kw2020202090@ubuntu: ~$
```

5. 실행파일 실행



The screenshot shows a terminal window titled "Ubuntu 64-bit - VMware Workstation 17 Player (Non-commercial use only)". The terminal is running on a system with the username "kw2020202090" and the hostname "ubuntu". The user has executed the command `./hello`, which has produced the following output:

```
kw2020202090@ubuntu:~$ ./hello
2020202090
Minseok Choi
Kwangwoon University
kw2020202090@ubuntu:~$
```

The terminal window also displays the date and time as "Mar 21 05:14". The left sidebar of the terminal shows various application icons, including the Dash icon, Home icon, Files icon, Applications icon, and the Dash icon.